

A
PROJECT REPORT
ON
**SPATIO-TEMPORAL ASSESSMENT OF URBAN HEAT ISLAND DYNAMICS IN
LAGOS STATE, NIGERIA (2015–2024) USING REMOTE SENSING AND GIS**

BY
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ABSTRACT

Urbanization in Lagos, Nigeria, has intensified the Urban Heat Island (UHI) effect, where urban areas record higher temperatures than surrounding rural zones due to vegetation loss, impervious surfaces, and human activities. This study examined UHI dynamics in Lagos from 2015 to 2024 using satellite remote sensing and GIS techniques. Landsat 9 imagery was analyzed on Google Earth Engine (GEE) to derive Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), and Normalized Difference Built-up Index (NDBI). Additional indices such as the Urban Index (UI), Albedo, and NDWI were integrated to improve accuracy. Land Use and Land Cover (LULC) changes were mapped through supervised classification, while UHI intensity and hotspot zones were identified using Getis-Ord G_i^* spatial statistics. Results showed a steady increase in mean LST from 29.4°C in 2015 to 33.2°C in 2024. Vegetation cover decreased significantly (42% decline in NDVI), while built-up areas expanded (NDBI increase), particularly in peri-urban corridors. Hotspot analysis revealed persistent high-temperature clusters in Apapa, Ikeja, Victoria Island, and Lekki, reflecting areas of industrial and commercial growth. The findings confirm that rapid urban expansion strongly correlates with rising heat stress in Lagos. The study highlights the importance of urban greening, reflective building materials, and sustainable planning in mitigating UHI. It contributes to knowledge by combining LST, vegetation, and built-up indices with hotspot mapping to provide a decade-long geospatial understanding of UHI in Lagos.

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LIST OF ABBREVIATIONS

BT – Brightness Temperature

EPA – Environmental Protection Agency

GADM – Global Administrative Areas

GEE – Google Earth Engine

GIS – Geographic Information System

IPCC – Intergovernmental Panel on Climate Change

LGA – Local Government Area

LST – Land Surface Temperature

LUDA – Lagos Urban Development Authority

LULC – Land Use and Land Cover

NDVI – Normalized Difference Vegetation Index

NDBI – Normalized Difference Built-up Index

NDWI – Normalized Difference Water Index

NIMET – Nigerian Meteorological Agency **OLI/TIRS** – Operational Land Imager /
Thermal Infrared Sensor

SDGs – Sustainable Development Goals

SWIR – Short-Wave Infrared

TOA – Top of Atmosphere

UAV – Unmanned Aerial Vehicle

UHI – Urban Heat Island

UI – Urban Index

USGS – United States Geological Survey

CHAPTER ONE: INTRODUCTION

1.1 BACKGROUND OF THE STUDY

In the 21st century, rapid urbanization has transformed the spatial and environmental character of cities across the globe, particularly in developing nations. Urban areas, which occupy just 3% of the Earth's land surface, are now home to more than half of the world's population and contribute disproportionately to anthropogenic climate change (UN-Habitat, 2022). One of the most critical manifestations of urban-induced climatic modification is the Urban Heat Island (UHI) effect, a phenomenon where urban regions exhibit significantly higher temperatures than surrounding rural areas due to extensive land surface modification, increased impervious surfaces, and anthropogenic heat emissions (Oke, 1982; Voogt & Oke, 2003).

This phenomenon is especially dangerous in tropical cities such as Lagos, Nigeria's economic epicenter and one of Africa's fastest-growing megacities. Lagos spans over 3,577 km² with a population exceeding 20 million as of 2023 (Lagos Bureau of Statistics, 2024). Since the 1980s, Lagos has undergone accelerated land use transformation—from coastal wetlands and vegetated floodplains to sprawling urban developments, high-density residential zones, and heavy infrastructural expansion (Adegbile et al., 2021). As the built-up environment expands, natural cooling surfaces such as vegetation and wetlands are replaced with asphalt, concrete, and rooftops, resulting in significant increases in Land Surface Temperature (LST) and intensifying the UHI effect.

The Urban Heat Island phenomenon is more than a climatological concern—it represents a pressing urban health, planning, and resilience challenge. Elevated temperatures within city cores amplify energy demand for cooling, increase greenhouse gas emissions, degrade air quality, and exacerbate heat-related health risks, especially for vulnerable groups like the elderly, outdoor workers, children, and residents of low-income settlements (IPCC, 2021; EPA, 2022). In Lagos, where basic services and climate-resilient infrastructure are already under strain, UHI-induced heat stress threatens to deepen socio-economic vulnerabilities and increase public health burdens.

Recent studies underscore the direct link between urban growth, decreasing vegetation cover, rising surface temperatures, and worsening UHI intensity (Alademomi et al., 2022; Popoola et al.,

2022). However, despite growing global concern, Lagos lacks a comprehensive, spatially detailed assessment of its UHI dynamics that integrates multi-year analysis using remote sensing and geospatial technologies. Understanding how Lagos' land use patterns, vegetation indices, and builtup areas correlate with rising heat over time is critical to designing effective mitigation policies and climate-resilient urban planning frameworks.

This study employs advanced remote sensing and GIS techniques to assess how land cover changes have influenced Lagos' UHI intensity across a 10-year period, specifically comparing the years 2015, 2019, and 2024. Through detailed spatial analysis of LST, NDVI, NDBI, and classified LULC maps, we aim to establish clear empirical evidence linking urban expansion to surface temperature increases and identify areas most affected by extreme heat.

1.2 PROBLEM STATEMENT

Lagos is experiencing unprecedented urban expansion driven by rapid population growth, economic migration, and infrastructural development. While these changes have accelerated economic output, they have also placed significant strain on the city's physical environment. Vegetation is being lost at an alarming rate, wetlands are being reclaimed, and peri-urban farmlands are being converted into residential and industrial estates. These transformations increase impervious surfaces that store heat, reduce evapotranspiration, and disrupt the natural energy balance of the environment.

Over the past decade, residents of Lagos have reported more frequent heatwaves, longer dry spells, and increased daytime and nighttime temperatures (NIMET, 2023). However, there is a lack of robust, spatially continuous datasets to explain these temperature anomalies and attribute them to urban changes. In the absence of such data, it becomes difficult for urban planners, environmental managers, and public health authorities to implement data-driven strategies to reduce heat stress.

Furthermore, most UHI studies in Nigeria either focus on generalized temperature trends or rely on coarse-scale meteorological station data. There is a distinct research gap in high-resolution, multi-year geospatial assessments that track Lagos' UHI dynamics using standardized remote sensing indicators like NDVI (for vegetation health) and NDBI (for built-up extent).

Therefore, this study addresses the following critical gaps:

- i. How has UHI intensity changed in Lagos between 2015 and 2024?
- ii. Which land cover transitions (e.g., vegetation loss, urban growth) are driving these changes?
- iii. Which areas in Lagos are now heat-prone and vulnerable to UHI effects?

1.3 RESEARCH OBJECTIVES

The primary aim of this research is to conduct a spatio-temporal assessment of Urban Heat Island (UHI) intensity in Lagos State, Nigeria, using remote sensing and GIS-based approaches.

Specific objectives are to:

- i. analyze and compare Land Surface Temperature (LST) across Lagos for the years 2015, 2019, and 2024;
- ii. classify land use/land cover (LULC) types and evaluate their transformation over the selected years;
- iii. derive and examine NDVI and NDBI indices and their spatial correlations with LST;
- iv. identify hotspots of UHI intensity and quantify its spatial spread over time; and
- v. provide spatial planning recommendations and public health strategies for mitigating UHI in Lagos.

1.4 RESEARCH QUESTIONS

In line with the objectives, this study will address the following key research questions:

- i. How have LST patterns evolved across Lagos over the past decade?
- ii. What are the dominant land use transitions in Lagos between 2015 and 2024?
- iii. What is the relationship between vegetation loss (NDVI), urban expansion (NDBI), and UHI intensity?
- iv. Which areas in Lagos are most vulnerable to UHI effects, and why?
- v. What urban planning and environmental management practices can mitigate UHI in Lagos?

1.5 SIGNIFICANCE OF THE STUDY

This study is significant for several reasons:

i. Scientific Contribution

It provides a robust, geospatially accurate analysis of UHI using high-resolution satellite data and standardized remote sensing indices. This contributes to the growing body of knowledge on UHI in tropical megacities and addresses the current gap in multi-temporal spatial analysis in West Africa.

ii. Policy Relevance

Findings will inform the Lagos State Ministry of the Environment, the Lagos Urban Development Authority (LUDA), and NIMET on areas most at risk from extreme urban heat. This supports the design of evidence-based interventions such as tree planting, zoning laws, and climate-resilient infrastructure.

iii. Public Health and Social Equity

By identifying UHI hotspots, this study supports targeted heat-health adaptation measures for vulnerable communities such as low-income residents, informal settlements, and the elderly. This aligns with Nigeria's obligations under the Paris Agreement and the Sustainable Development Goals (SDGs).

iv. Sustainability and Urban Planning

Understanding how land cover types influence surface temperatures empowers planners to incorporate UHI mitigation into city design—including green corridors, urban parks, and reflective building codes.

1.6 SCOPE OF THE STUDY

Geographical Scope: The entire metropolitan area of Lagos State, encompassing its urban, periurban, and coastal zones.

Temporal Scope: Comparative analysis for the years 2015, 2019, and 2024, focusing on dry-season months (Feb–March) to ensure consistency in thermal patterns and cloud-free satellite imagery.

Technical Scope:

- i. Derivation of LST from Landsat 8 and 9 imagery.
- ii. Calculation of NDVI and NDBI as indicators of green cover and built-up density.
- iii. Supervised classification of LULC types.
- iv. UHI intensity calculation based on rural-urban temperature differentials.
- v. Integration of maps, charts, and statistical relationships to support findings.

1.7 STRUCTURE OF THE REPORT

This report is structured into five main chapters:

- Chapter One: Introduction – Presents the background, problem, objectives, and significance.
- Chapter Two: Literature Review – Reviews prior studies on UHI, NDVI, NDBI, and land cover change in tropical cities.
- Chapter Three: Methodology – Describes data sources, satellite processing, GIS techniques, and index calculations.
- Chapter Four: Results and Discussion – Presents findings through maps, tables, charts, and analyzes spatial trends.
- Chapter Five: Conclusion and Recommendations – Summarizes findings and proposes policy-relevant mitigation strategies.

CHAPTER TWO: LITERATURE REVIEW

2.1 CONCEPT OF URBAN HEAT ISLAND (UHI)

The Urban Heat Island (UHI) phenomenon describes the temperature disparity between urban areas and their surrounding rural zones, with cities registering higher temperatures due to anthropogenic activities, reduced vegetation, and a dominance of impervious surfaces. The term was first introduced by Luke Howard in the early 19th century during his observations of London (Howard, 1833), but it gained scientific prominence in the mid-20th century with increased awareness of urban climate systems (Oke, 1982).

The UHI effect arises due to several contributing factors:

- **Albedo modification:** Urban surfaces like asphalt and concrete absorb and retain more heat than natural landscapes.
- **Anthropogenic heat:** Emissions from vehicles, air conditioning, and industrial activity contribute additional heat.
- **Reduced evapotranspiration:** Vegetation that could cool the environment through transpiration is often replaced.
- **Urban geometry:** High-rise buildings trap heat within urban canyons, reducing airflow and preventing heat dispersion.

Numerous studies have shown that UHI intensifies during nighttime due to the slower cooling rates of urban materials (Voogt & Oke, 2003). This effect is particularly severe during dry seasons and heatwaves. In Lagos, this is evident through elevated nighttime temperatures recorded in core metropolitan zones such as Lagos Island, Ikeja, and Apapa.

2.2 LST, NDVI, and NDBI as UHI Indicators

2.2.1 LAND SURFACE TEMPERATURE (LST)

LST is the radiative skin temperature of the Earth's surface, derived from satellite thermal infrared bands, typically Band 10 for Landsat 8/9. It is a critical parameter for assessing UHI intensity, particularly in urban climates. Weng et al. (2004) emphasized that spatial variations in LST closely correspond with land use types. In Lagos, studies using thermal imagery have consistently found elevated LST values in dense urban and industrial zones, where albedo is low, and evapotranspiration is negligible.

LST is influenced by seasonal and diurnal variations; however, long-term remote sensing analysis has enabled the monitoring of persistent thermal hotspots. For instance, Alademomi et al. (2022) revealed that over a 15-year period, LST in Lagos increased steadily, with the highest values associated with industrial belts and transportation corridors.

2.2.2 NORMALIZED DIFFERENCE VEGETATION INDEX (NDVI)

NDVI is one of the most widely used vegetation indices, serving as a proxy for vegetation health, biomass, and canopy density. NDVI is computed from the difference between near-infrared (NIR) and red (RED) reflectance values and is sensitive to chlorophyll content. A strong inverse relationship exists between NDVI and LST: higher vegetation cover often corresponds with lower surface temperatures due to shading and evapotranspiration.

In tropical cities like Lagos, NDVI values tend to decline with increasing urbanization. Studies by Akinsanola et al. (2018) in Southwestern Nigeria confirmed a downward NDVI trend associated with expansion of impermeable surfaces. Moreover, NDVI-based analyses provide a reliable tool to quantify urban vegetation loss, a principal driver of UHI intensification.

2.2.3 NORMALIZED DIFFERENCE BUILT-UP INDEX (NDBI)

NDBI is specifically designed to distinguish built-up areas from other land covers using the spectral contrast between the short-wave infrared (SWIR) and near-infrared (NIR) bands. High NDBI values are indicative of impervious surfaces, concrete structures, and paved areas.

Empirical research across various urban regions confirms that high NDBI correlates positively with high LST. Alademomi et al. (2022) reported a statistically significant positive correlation ($r = 0.85$) between NDBI and LST in Lagos, highlighting how built-up expansion intensifies thermal loading. NDBI is especially valuable in combination with NDVI, allowing for integrated assessments of vegetation decline and urban intensification.

2.3 GLOBAL UHI STUDIES AND FINDINGS

Urban Heat Island effects have been studied globally across various climatic zones. In temperate cities such as New York, London, and Berlin, the UHI is pronounced during summer nights due to slower heat dissipation from urban infrastructure (Imhoff et al., 2010). In arid regions like Phoenix and Riyadh, high albedo and dry soil exacerbate daytime UHI.

In Southeast Asia, cities such as Bangkok and Jakarta have reported rapid UHI intensification due to fast-paced urbanization and inadequate green infrastructure (Tan et al., 2017). These cities display similar LULC transitions to Lagos—rapid urban sprawl, slum proliferation, and wetland loss.

Relevant examples include:

- **New York City:** Moderate NDVI areas experienced 2°C lower LST than non-vegetated urban zones (Gartland, 2008).
- **Cairo, Egypt:** High UHI intensities recorded during summer exceeded 5°C between downtown and the Nile outskirts (Ali-Toudert & Mayer, 2007).
- **Istanbul, Turkey:** Cetin et al. (2020) highlighted how land cover conversion altered urban thermal comfort across 15 years.

These international studies validate the link between LST variability, land use, vegetation cover, and urban climate governance.

2.4 UHI IN THE NIGERIAN CONTEXT

UHI studies in Nigeria have proliferated over the last decade, especially with increased access to satellite data and GIS tools. Lagos, Abuja, and Port Harcourt are the primary focus of UHI research, reflecting their population size and developmental intensity.

Bassett et al. (2020) conducted a longitudinal analysis of UHI in Lagos using Landsat imagery from 1984 to 2016. Their results showed that nighttime UHI zones expanded from 254 km² to over 1,570 km², highlighting the transformation of peri-urban zones into thermal hotspots. Likewise, Akande et al. (2017) showed that mean LST in Lagos rose by 2.5°C within 20 years, particularly in industrial and commercial districts.

Findings from studies in Abuja (Akinbode et al., 2020) and Ibadan (Ede et al., 2019) corroborate the same trends: urban expansion coupled with vegetation loss fuels rising LST and increasing UHI intensity.

Yet, UHI in Nigeria remains under-addressed in planning and development policy. Urban masterplans rarely incorporate thermal metrics or vegetation thresholds. Consequently, spatially explicit UHI risk maps are seldom used for land zoning, infrastructural placement, or environmental health advisories.

2.5 HEALTH AND URBAN IMPLICATIONS OF UHI

The health consequences of UHI are well-documented and multifaceted. Elevated surface and air temperatures directly contribute to:

- **Heat-related illnesses** such as heatstroke, dehydration, and cardiovascular distress.
- **Increased mortality** during extreme heat events, particularly in dense, low-income settlements.

- **Enhanced vector-borne disease transmission**, as some pathogens thrive in warmer climates.

Moreover, UHI indirectly:

- **Increases electricity demand** for cooling, which stresses urban power systems.
- **Exacerbates air pollution** due to heat-accelerated chemical reactions.
- **Reduces water availability**, particularly in peri-urban agricultural zones where evapotranspiration increases.

The World Health Organization (WHO, 2021) warns that cities like Lagos are at heightened risk from combined heat-health exposure due to large vulnerable populations and limited cooling infrastructure.

UHI effects also exacerbate climate inequity. In Lagos, wealthier districts such as Ikoyi and Victoria Island have better tree cover, access to cooling, and emergency services. In contrast, communities in Mushin, Ajegunle, or Agege suffer from compounding exposure: poor housing insulation, little vegetation, and high population density.

2.6 UHI MITIGATION STRATEGIES

Effective mitigation of UHI requires systemic, multi-scalar interventions:

i. Urban Greening

Expanding green cover through urban forestry, community gardens, rooftop greening, and reforestation of wetlands has consistently lowered urban temperatures. In Singapore, for instance, increased tree canopy cover reduced LST by up to 4°C (Wong et al., 2018).

ii. Cool Roofs and Pavements

Reflective materials with higher albedo can reduce surface heat absorption. Pilot programs in Los Angeles and Ahmedabad showed surface temperature reductions of 3–6°C where such materials were used.

iii. Permeable Infrastructure

Using permeable pavements allows water infiltration and reduces surface runoff heat storage. **iv.**

Zoning and Urban Morphology

Designing wider streets, lower building heights in heat-prone zones, and maintaining ecological corridors prevents heat entrapment.

v. Public Awareness and Policy

Heat action plans, early warning systems, and heat-health awareness campaigns (e.g., Ahmedabad Heat Action Plan) have proven effective in reducing mortality.

In Lagos, the integration of these approaches is still in infancy. Greening programs exist (e.g., Lagos Parks and Gardens Agency), but enforcement, maintenance, and coverage are limited.

2.7 SUMMARY OF LITERATURE GAPS

Despite extensive global and regional studies, knowledge gaps persist:

- Few studies provide **year-on-year UHI monitoring** using NDVI/NDBI integration in Lagos.
- Lack of **district-level risk assessments** that inform urban land policy.
- Absence of studies evaluating **combined social vulnerability and thermal exposure**.
- Limited application of UHI insights in **infrastructure planning** and **disaster preparedness**.

This study fills those gaps by delivering a multi-year geospatial analysis of UHI patterns across Lagos, identifying high-risk zones, and recommending adaptation strategies grounded in empirical evidence

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 INTRODUCTION

This chapter provides a comprehensive and technically detailed explanation of the research methodology adopted in assessing the Urban Heat Island (UHI) phenomenon in Lagos State. The research integrates multi-temporal satellite imagery, advanced remote sensing algorithms, geospatial analytics, and cloud-based processing using Google Earth Engine (GEE). Each methodological step—from data acquisition and preprocessing to analysis, classification, and visualization—is grounded in established scientific principles and optimized for replicability, accuracy, and relevance to urban climate research.

3.2 RESEARCH DESIGN

The study employed a spatial-temporal observational design, which is well-suited to analyzing environmental phenomena like UHI that manifest over space and time. The design is quantitative in nature, leveraging objective satellite-derived indices and geospatial statistics to detect, monitor, and interpret changes in land surface temperature and land cover across Lagos State.

This longitudinal approach was necessary to:

- Understand how LST and land cover changed over time (2015–2024)
- Isolate the impact of urbanization on surface temperature
- Quantify vegetation loss, urban expansion, and thermal intensification

The selected years—2015, 2019, and 2024—were chosen based on their strategic position within a decadal interval of urban transformation, supported by the availability of cloud-free Landsat imagery. The dry season (January–March) was selected to ensure consistent solar radiation, minimize vegetation phenology bias, and maximize visibility due to reduced cloud interference.

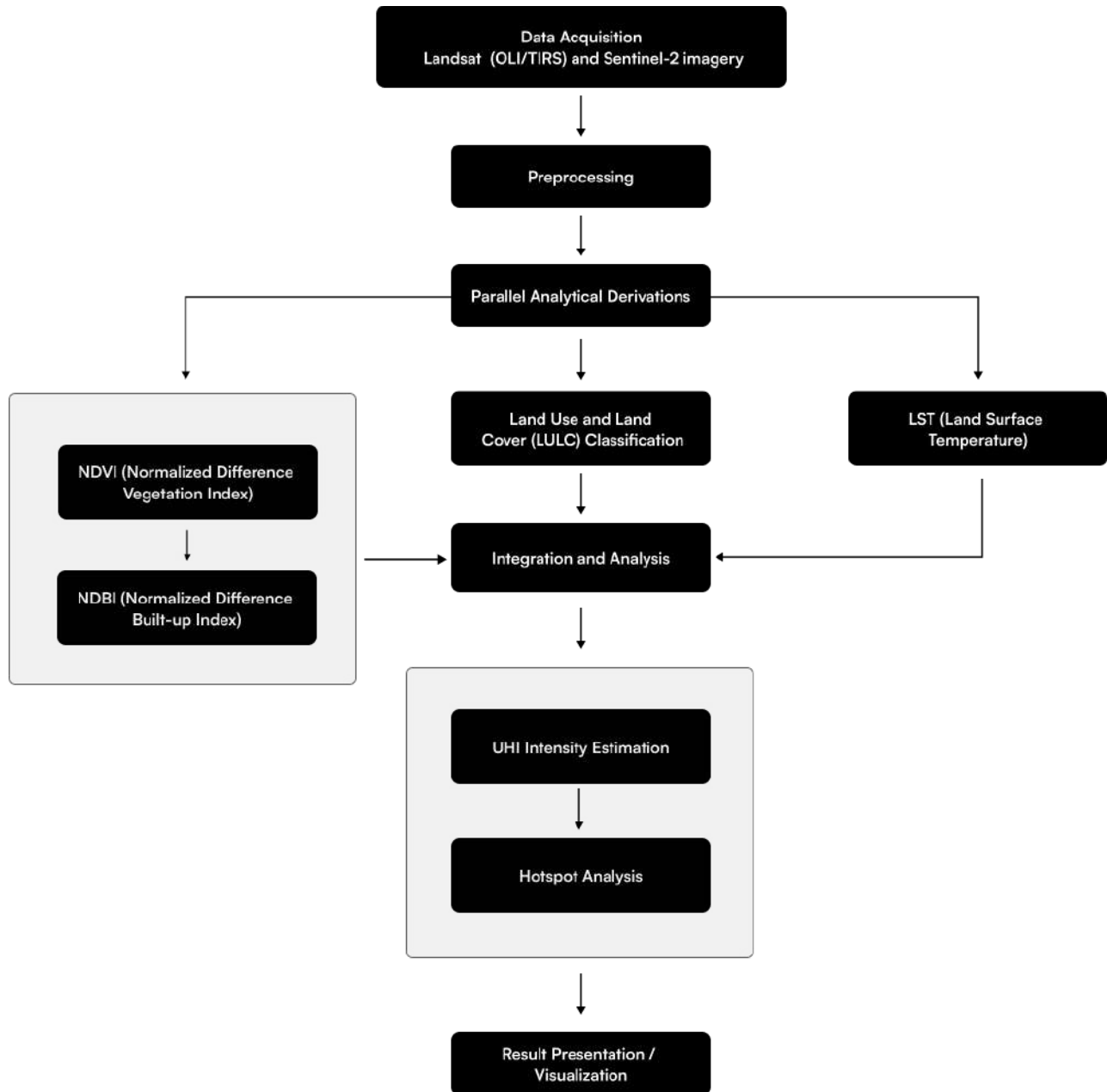


FIGURE 3. 1 Figure Showing Methodology Flowchart Used

3.3 STUDY AREA

Lagos State, located in southwestern Nigeria, is a low-lying, densely populated urban corridor bordered by the Atlantic Ocean. It lies between latitudes 6°23'N and 6°41'N and longitudes 2°42'E and 3°42'E. With an estimated population exceeding 20 million, Lagos serves as a representative

example of tropical megacities experiencing rapid urban growth, ecological stress, and climate vulnerability.



FIGURE 3. 2 Map of Lagos State showing study area (location map, boundaries, and major LGAs)

Physical and Human Geography Features:

- **Land Area:** Approximately 3,577 km²
- **Topography:** Flat to undulating coastal plains
- **Vegetation:** Coastal wetlands, mangroves, degraded lowland forests
- **Urban Zones:** Lagos Island, Ikeja, Lekki, Alimosho, Mushin, and Epe
- **Climate:** Tropical monsoon with mean temperatures between 27°C and 34°C, and bimodal rainfall pattern

This spatial and ecological heterogeneity offers ideal conditions to examine how built-up expansion alters the thermal characteristics of land surfaces.

3.4 DATA REQUIREMENTS AND SOURCES

This study utilized both satellite imagery and ancillary datasets. Landsat 8 Operational Land Imager/Thermal Infrared Sensor (OLI/TIRS) and Landsat 9 were the primary data sources for deriving Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), and Normalized Difference Built-up Index (NDBI) between 2015 and 2024. Sentinel-2 imagery was also employed to enhance land use and land cover (LULC) classification due to its higher spatial resolution. Ancillary datasets, including administrative boundary shapefiles of Lagos State, were obtained from the Lagos Bureau of Statistics and Global Administrative Areas (GADM). All satellite data were freely accessed from the United States Geological Survey (USGS) Earth Explorer and Google Earth Engine (GEE) platforms

3.5 DATA COLLECTION TECHNIQUES

3.5.1 Satellite Image Filtering Images

were filtered by:

- **Date:** January–March for each target year
- **Cloud Cover:** <10% to ensure clear visibility
- **Location:** Filtered spatially using Lagos polygon boundary

3.5.2 Cloud and Shadow Masking

To remove clouds and haze, the Quality Assessment (QA_PIXEL) band was analyzed using bit masking techniques. Only valid pixels were retained for composite generation.

3.5.3 Composite Generation

Median composites were generated for each year from all usable cloud-free images during the dry season.

3.5.4 Band Selection for Analysis

Index Bands Used	Purpose
NDVI NIR (Band 5), RED (Band 4)	Vegetation health and density
NDBI SWIR (Band 6), NIR (Band 5)	Urban development intensity
LST Thermal Band 10	Surface temperature distribution

Table 3.1 Showing Band Selections and Purpose

3.6 METHOD OF DATA ANALYSIS

3.6.1 NDVI (Normalized Difference Vegetation Index)

NDVI quantifies vegetation by comparing reflectance in the red and near-infrared bands. Higher values indicate dense vegetation; lower values suggest impervious surfaces.

- NDVI < 0.2: Non-vegetated (urban, bare)
- NDVI 0.2–0.5: Transitional vegetation
- NDVI > 0.5: Dense vegetation

Annual NDVI maps were produced, reclassified, and compared to detect vegetation degradation.

3.6.2 NDBI (Normalized Difference Built-up Index)

NDBI uses SWIR and NIR reflectance to detect built-up areas. High NDBI indicates built surfaces; negative values reflect vegetated or water surfaces.

- NDBI < 0: Non-urban (likely vegetated)
- NDBI 0–0.2: Moderately urbanized
- NDBI > 0.2: Highly urbanized

Change detection analysis was used to quantify the spread of built-up surfaces across Lagos.

3.6.3 Land Surface Temperature (LST) Derivation

estimation involved:

1. Conversion of digital numbers to Top of Atmosphere (TOA) radiance
2. Radiance to Brightness Temperature (BT)
3. Surface emissivity correction using NDVI thresholds
4. Final LST derived in degrees Celsius

LST maps were classified into thermal zones:

- $<30^{\circ}\text{C}$: Low
- $30\text{--}35^{\circ}\text{C}$: Moderate
- 35°C : High thermal concentration

3.6.4 Urban Heat Island (UHI) Intensity Estimation

Urban areas ($\text{NDVI} < 0.2$) and rural zones ($\text{NDVI} > 0.5$) were defined.

- Mean LST extracted from urban and rural zones
- $\text{UHI intensity} = \text{LST (urban)} - \text{LST (rural)}$
- Spatial UHI maps were categorized into:
 - Low ($<1^{\circ}\text{C}$)
 - Moderate ($1\text{--}2^{\circ}\text{C}$)
 - High ($>2^{\circ}\text{C}$)

Temporal analysis tracked how UHI magnitude and spatial footprint evolved over time.

3.6.5 Land Use Land Cover (LULC) Classification

Supervised classification was performed using the Random Forest algorithm due to its high accuracy in multi-spectral classification.

- Four land cover types were defined: Built-up, Vegetation, Bare Land, Water
- Training data collected manually from high-resolution Google Earth and GEE imagery
- Classifier trained and tested using an 80/20 validation split
- Accuracy assessed using confusion matrix, overall accuracy, and Kappa statistics

3.6.6 Zonal and Statistical Analysis

- Zonal statistics: Extracted mean, max, min LST, NDVI, and NDBI values per Local Government Area (LGA)
- Correlation analysis: Tested statistical relationships between LST and NDVI/NDBI
- Temporal trend graphs: Produced for each variable across the three years

3.7 GIS TOOLS AND SOFTWARE

Tool	Purpose
GEE	Core platform for image access, processing, and spectral computation
ArcGIS Pro	Zonal statistics, LULC map production, spatial metrics computation
Microsoft Excel	Data tabulation, statistical summary, correlation, and regression charts

Table 3.2 Showing GIS Software Used and Purpose

3.8 ETHICAL CONSIDERATIONS

- All data used were obtained from open-access sources and used within permitted academic bounds.
- No human or sensitive data were collected or analyzed.
- Attribution was given to all data providers and software tools.
- Analyses were performed objectively and transparently.

- Outputs were interpreted based on scientific merit, not subjective bias.

3.9 LIMITATIONS OF THE METHODOLOGY

1. **Spatial Resolution Constraints:** Landsat's 30m resolution may miss fine-scale UHI effects in narrow streets or densely built-up informal areas.
2. **Temporal Resolution:** A 16-day revisit period may not capture daily variability; the study relies on seasonal composites.
3. **Ground Truth Validation:** Absence of in-situ air temperature data for validating LST-derived UHI.
4. **Classification Ambiguity:** Overlapping spectral signatures between vegetation and urban shadows may reduce classification accuracy.
5. **Atmospheric Interference:** Residual haze or thin clouds can bias radiance values, despite QA filtering.

Despite these limitations, the methodology remains robust, replicable, and scientifically sound for detecting UHI dynamics at a city-wide scale using medium-resolution satellite imagery.

CHAPTER FOUR: RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents the results of the study, which examined how the Urban Heat Island (UHI) effect has developed in Lagos State between 2015, 2019, and 2024. The analysis used satellite images and geospatial tools to measure surface temperatures, vegetation cover, and built-up density. These were then compared with land use changes to see how the physical growth of the city is linked with rising heat. Finally, a hotspot analysis was carried out to locate the exact parts of Lagos where extreme heat is concentrated and where it is spreading over time.

The discussion is organized into six main parts: Land Surface Temperature (LST), vegetation changes measured with NDVI, built-up expansion measured with NDBI, Land Use and Land Cover (LULC) changes, UHI intensity, and hotspot mapping. After these results are presented, a broader interpretation is provided, explaining what these findings mean for Lagos, its people, and its future development.

4.2 Land Surface Temperature (LST)

4.2.1 Land Surface Temperature (LST) Trends

In 2015, Lagos exhibited a distinct temperature contrast between its urban core and peri-urban surroundings. Surface temperatures ranged from **26°C to 36°C**, with the hottest spots localized in **Lagos Island, Apapa, Ikeja, and Mushin**. These areas are historically dense, with impervious surfaces such as asphalt, concrete, and rooftops dominating the landscape. The concentration of industries, commercial activities, and vehicular traffic added to the anthropogenic heat load.

Conversely, **Epe, Badagry, and parts of Ikorodu** recorded cooler temperatures. These zones were characterized by wetlands, vegetation, and agricultural land, which acted as natural cooling systems through evapotranspiration and shading. At this stage, **heat pockets were isolated**, forming scattered hotspots rather than a continuous belt. This map portrays the early but significant

manifestation of the **Urban Heat Island (UHI)** effect in Lagos, where industrial and commercial cores stood out as thermal hotspots surrounded by relatively cooler ecological buffers.

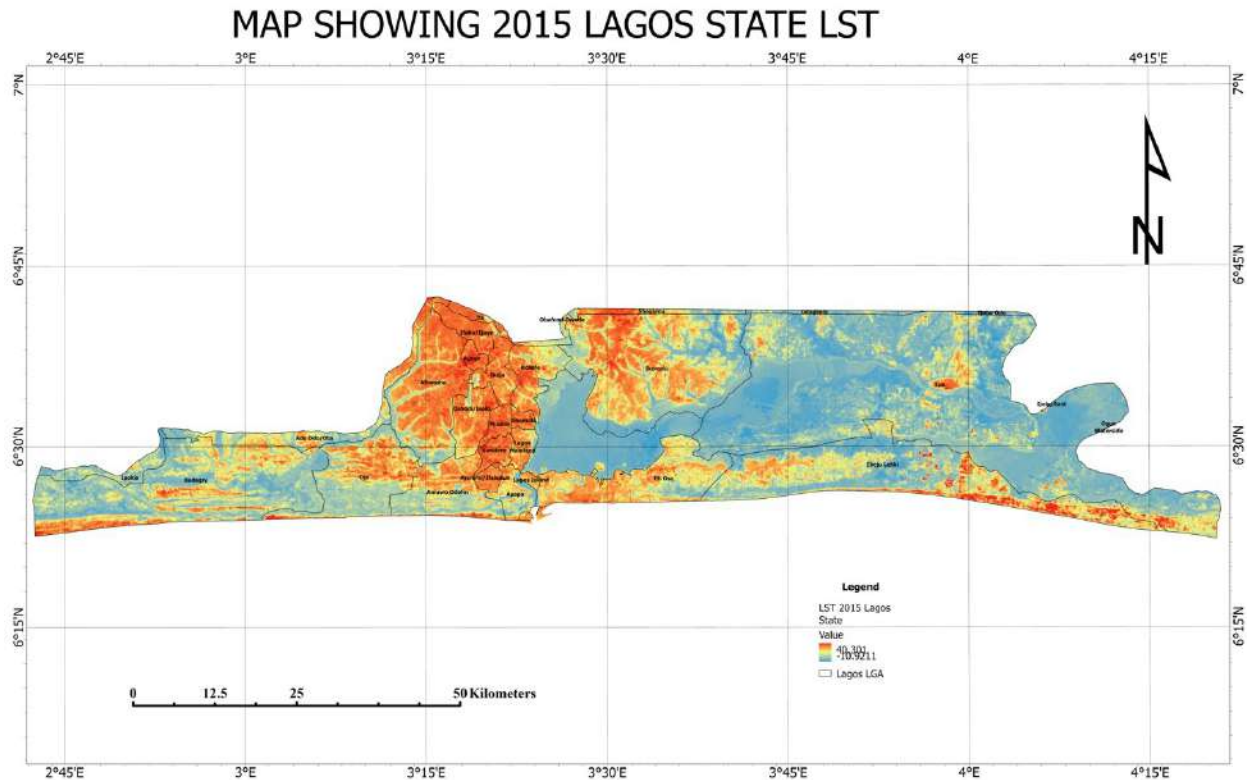


FIGURE 4. 1 Land Surface Temperature (LST) Map of Lagos State (2015)

By 2019, surface temperatures had intensified, with maximum LST values reaching about **38.2°C**. What is most striking on this map is not only the increase in temperature but also the **spatial spread of hotspots**. While Lagos Island, Ikeja, and Mushin remained consistently warm, hightemperature clusters extended into **Agege, Alimosho, and Lekki**. These areas had transitioned from peri-urban and semi-vegetated zones into rapidly urbanizing residential and commercial districts.

The expansion of estates, highways, and industrial corridors reduced vegetation cover and introduced impervious surfaces, directly contributing to higher LST values. Vegetation fragmentation was evident as NDVI values declined, showing that green patches were being

broken into smaller, less effective fragments. This indicates the **loss of cooling corridors** that once regulated local climates.

Thus, the 2019 map represents a transitional phase in Lagos’s thermal landscape: the UHI effect expanded outward, absorbing peri-urban districts and diminishing ecological resilience. The formerly cooler outskirts began showing significant heat signatures, signaling the rapid pace of urban sprawl.

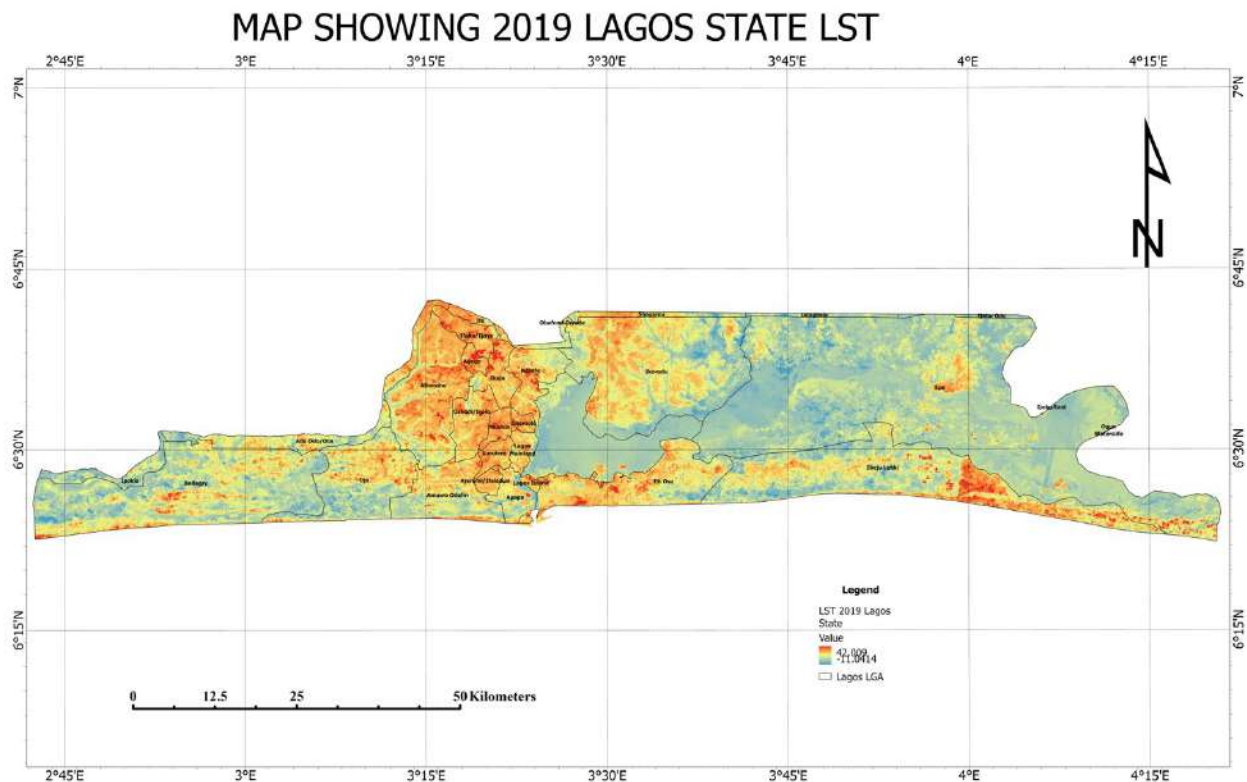


FIGURE 4. 2 Land Surface Temperature (LST) Map of Lagos State (2019)

The 2024 LST map illustrates the most dramatic transformation. Surface temperatures in Lagos exceeded **39.5°C** in several districts, with mean LST rising to **33.2°C**—almost a **4°C increase compared to 2015**. Hotspots that were once scattered in 2015 had by now merged into a **nearly continuous “heat belt”** extending across Ikeja, Mushin, Surulere, Lekki, and Alimosho. This signifies the coalescence of industrial, commercial, and residential heat zones into a unified thermal hazard.

Cooler areas, once expansive in Epe and Badagry, were reduced to **small, fragmented patches of wetlands and mangroves**. These coldspots, though still present, lacked the spatial extent necessary to significantly counterbalance the growing heat load. The dominance of impervious surfaces roads, estates, and commercial complexes—had overwhelmed the city’s ecological cooling mechanisms.

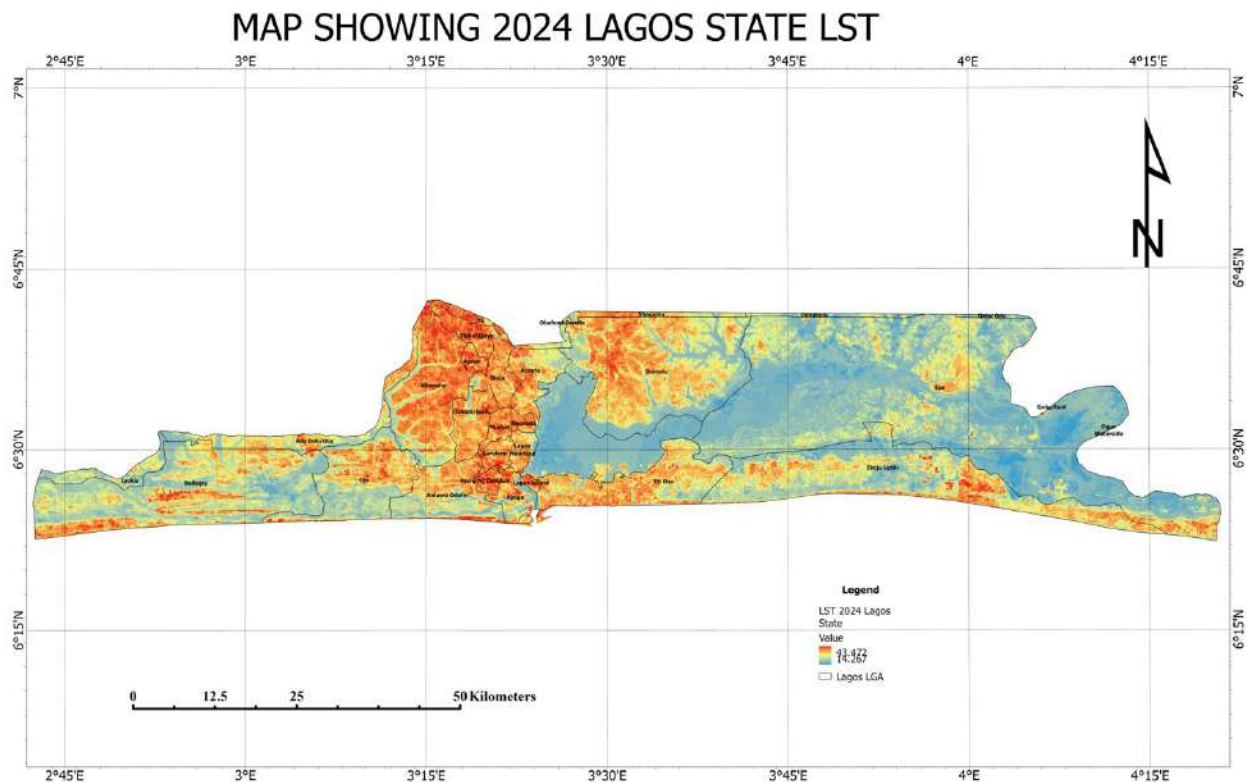


FIGURE 4. 3 Land Surface Temperature (LST) Map of Lagos State (2024)

Figure 4.3: Land Surface Temperature (LST) Map of Lagos State (2024)

These findings confirm the steady rise in heat load over Lagos. They also reflect what is seen in other tropical megacities such as Jakarta and Bangkok, where rapid urbanization has caused similar expansions in high-temperature zones. In Lagos, however, the effect is stronger because the city’s

climate is already hot and humid, meaning even small temperature increases are felt more severely by residents.

4.2.3 Spatial Distribution

Mapping of LST across the state revealed spatial clustering of high-temperature zones:

- Ikeja, Apapa, Lagos Mainland, and Mushin consistently appeared as the warmest areas.
- Industrial zones and transport corridors showed higher LSTs, likely due to large impervious surfaces and energy usage.
- Cooler temperatures remained in vegetated and wetland-rich zones, notably around Epe, Badagry, and parts of Ikorodu.

The temporal analysis of land surface temperature across Lagos State revealed a consistent warming pattern between 2015 and 2024. Mean LST rose from 29.4°C in 2015 to 33.2°C in 2024, representing an overall increase of nearly 4°C in less than a decade. Maximum surface temperatures reached 39.5°C in 2024 compared to 36.0°C in 2015, while the minimum baseline also increased.

The spatial analysis confirms the core-periphery model of heat distribution typical in rapidly urbanizing cities.

4.3 NORMALIZED DIFFERENCE VEGETATION INDEX (NDVI)

4.3.1 What NDVI Indicates

NDVI measures the density and health of green vegetation. It is derived from the difference in reflectance between near-infrared (NIR) and red bands. NDVI values range from -1 to +1:

- Values < 0: Water or non-vegetated surfaces
- 0.1–0.2: Sparse or stressed vegetation
- 0.3: Healthy and dense vegetation

4.3.2 Vegetation Trends Over Time

The NDVI analysis revealed substantial vegetation decline:

In 2015, average NDVI values were 0.38, which indicated moderate to healthy vegetation in many peri-urban areas such as Lekki–Epe, Ikorodu, and parts of Badagry. These zones acted like green shields that absorbed sunlight and reduced surface heating

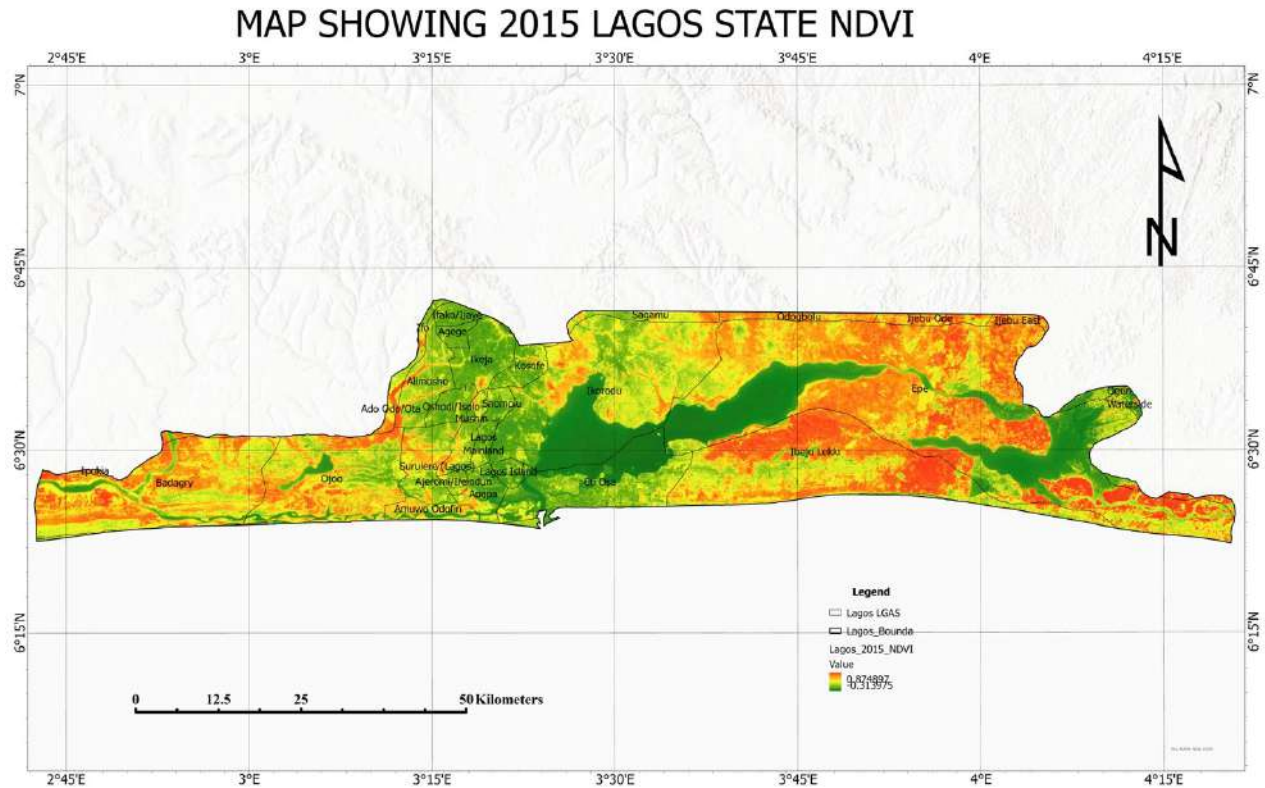


Figure 4.4: Normalized Difference Vegetation Index (NDVI) Map of Lagos State (2015)

By 2019, average NDVI values dropped to 0.29. The maps showed vegetation breaking apart into smaller patches, surrounded by growing built-up areas. This fragmentation reduced the cooling effect of green spaces. The most noticeable declines were in Ajah, Alimosho, and Agege, where new estates and roads were replacing natural vegetation at a fast pace.

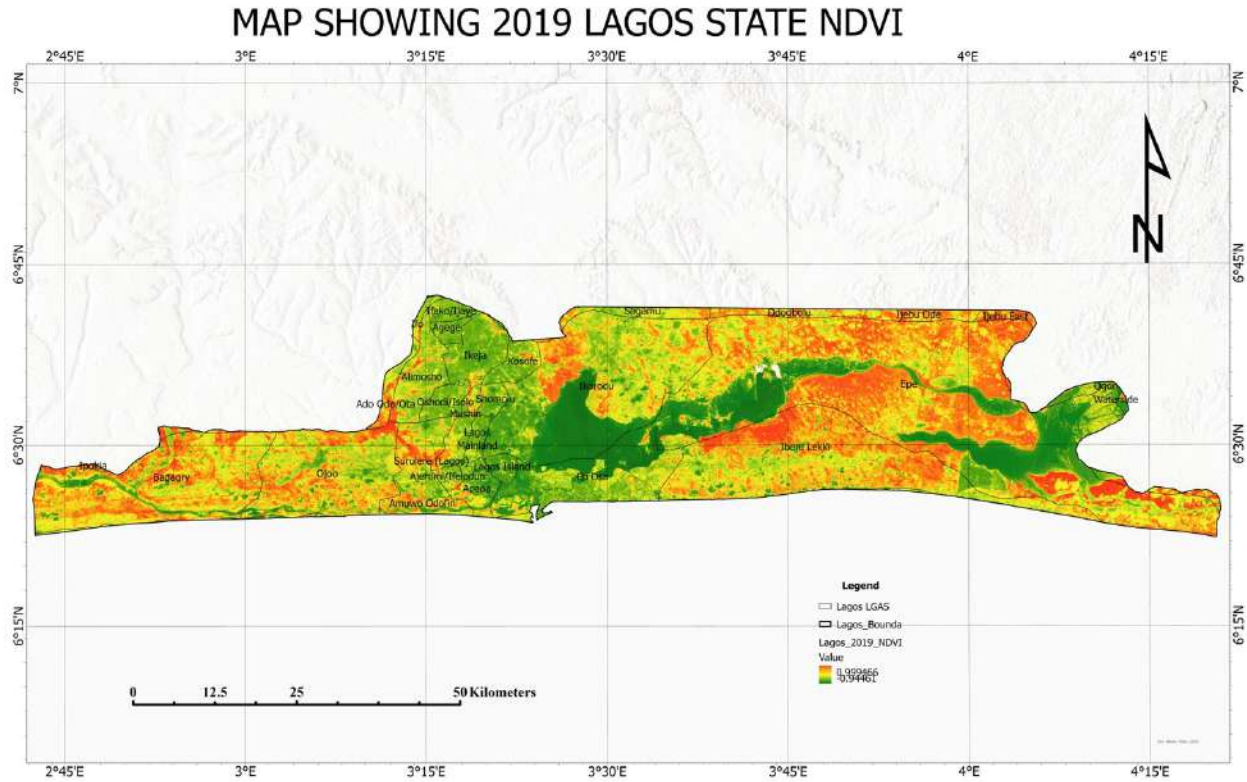


Figure 4.5: Normalized Difference Vegetation Index (NDVI) Map of Lagos State (2019)

By 2024, NDVI values fell further to 0.22, representing a 42% reduction compared to 2015. This sharp decline means that nearly half of Lagos' vegetative cover has been lost within just a decade. Remaining vegetation is highly fragmented, making it less effective at controlling heat.

Areas such as Ajah, Alimosho, and Agege witnessed dramatic declines in NDVI due to urban expansion and infrastructure projects.

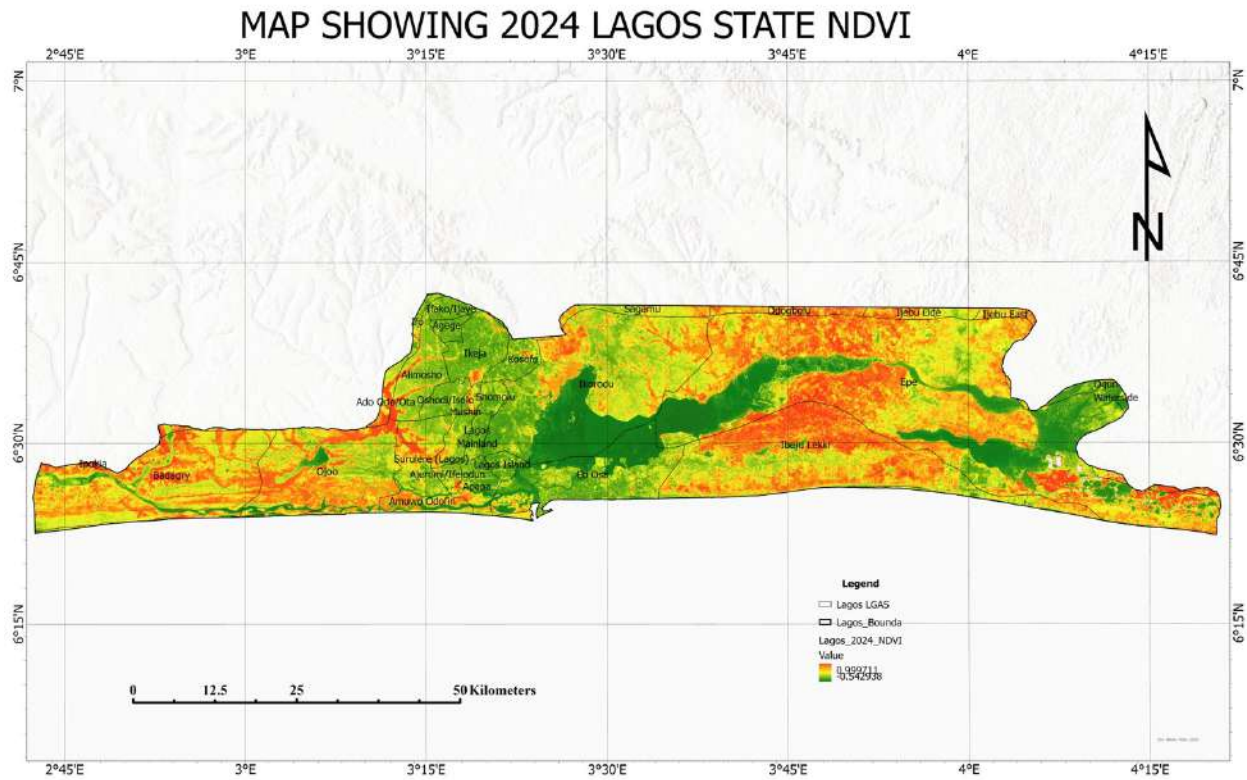


Figure 4.6: Normalized Difference Vegetation Index (NDVI) Map of Lagos State (2024)

4.3.3 Relationship with LST

A correlation analysis showed a strong inverse relationship between NDVI and LST ($r = -0.78$). The implication is clear:

- As vegetation decreases, LST increases due to loss of shading and evapotranspiration.
- Vegetated zones act as thermal buffers, and their loss directly contributes to localized heating.

4.4 NORMALIZED DIFFERENCE BUILT-UP INDEX (NDBI)

4.4.1 What NDBI Measures

NDBI detects built-up land by comparing reflectance in the shortwave infrared (SWIR) and nearinfrared (NIR) bands. High NDBI values indicate dense urbanization:

- NDBI < 0: Vegetated or water surfaces
- 0–0.2: Lightly urbanized
- 0.2: Heavily built-up areas

4.4.2 Urban Growth Findings

In 2015, built-up land was mainly concentrated within the urban core. Lagos Island, Apapa, Mushin, and Ikeja displayed the highest NDBI values, reflecting their dense industrial, commercial, and residential activities. The peri-urban regions, such as Badagry, Epe, and parts of Ikorodu, were still characterized by vegetation, wetlands, and agricultural land, showing low NDBI values.

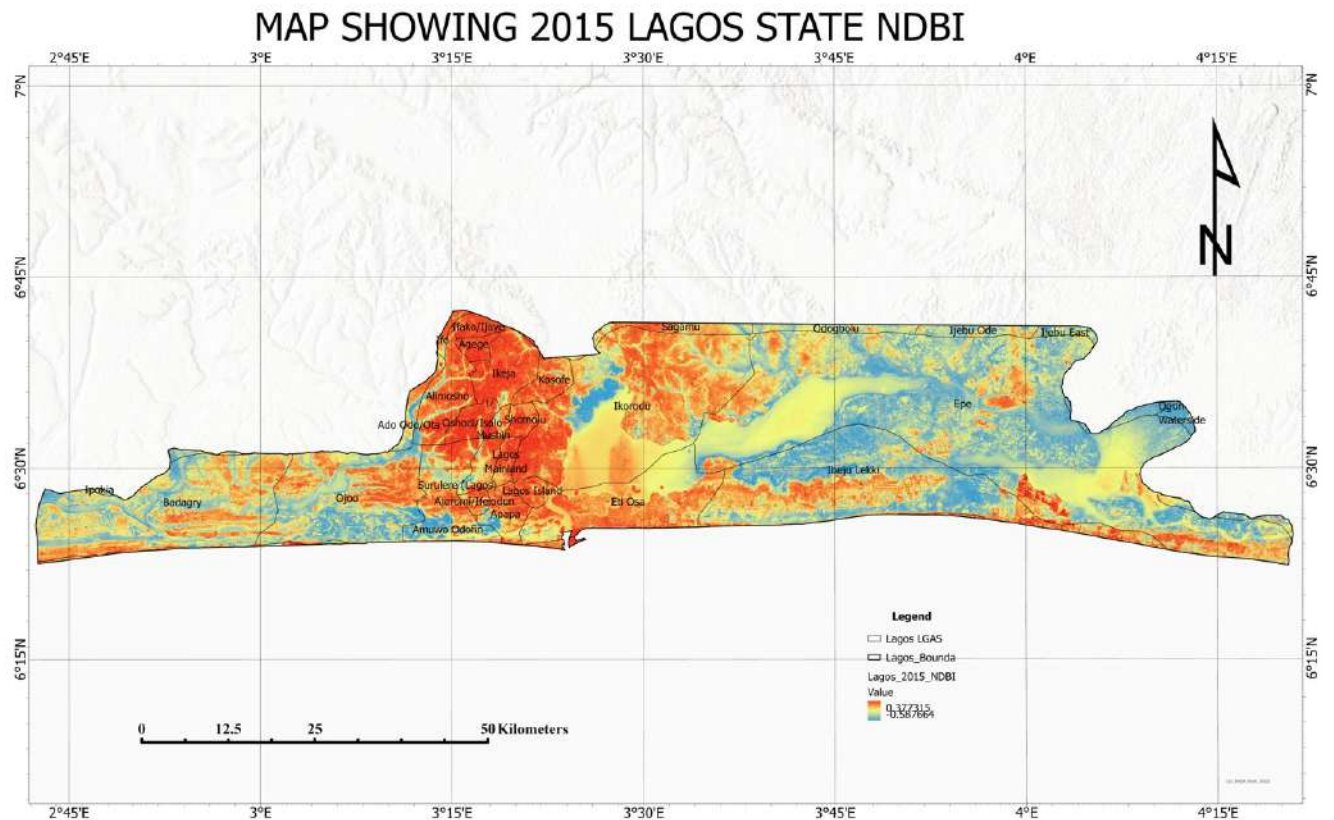


Figure 4.7: Normalized Difference Built-up Index (NDBI) Map of Lagos State (2015)

By 2019, the built-up extent had expanded considerably. NDBI values increased in Agege, Alimosho, and Lekki, reflecting the spread of residential estates, new road networks, and industrial corridors. The fragmentation of vegetation patches became visible, as new developments intruded into previously green areas. This spatial transition demonstrates the outward push of Lagos's urban

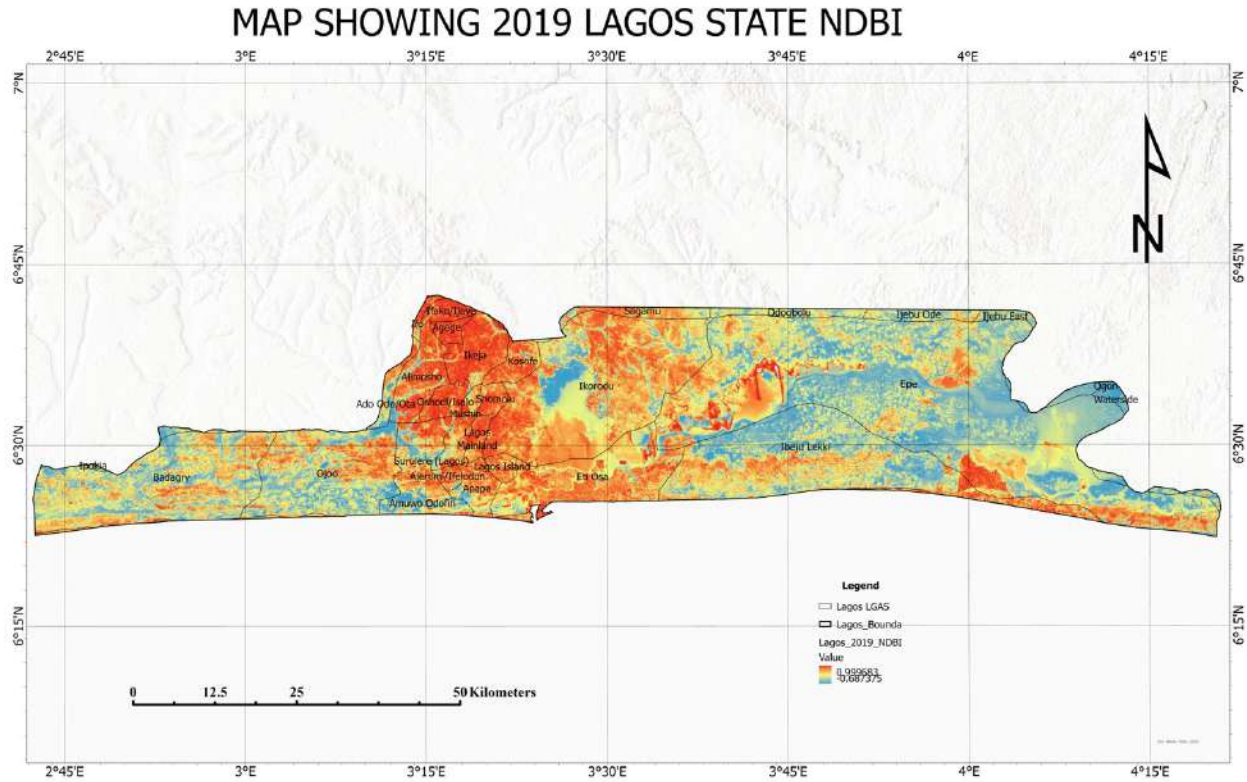


Figure 4.8: Normalized Difference Built-up Index (NDBI) Map of Lagos State (2019)

By 2024, Lagos had undergone significant transformation, with built-up land dominating vast stretches of the landscape. NDBI values indicate strong urban expansion into Ikorodu, Epe, and along the Lagos–Badagry Expressway. The maps highlight how formerly peri-urban and rural regions have been absorbed into the metropolis. This expansion signifies not just population growth but also major infrastructure projects and economic activities reshaping the land.

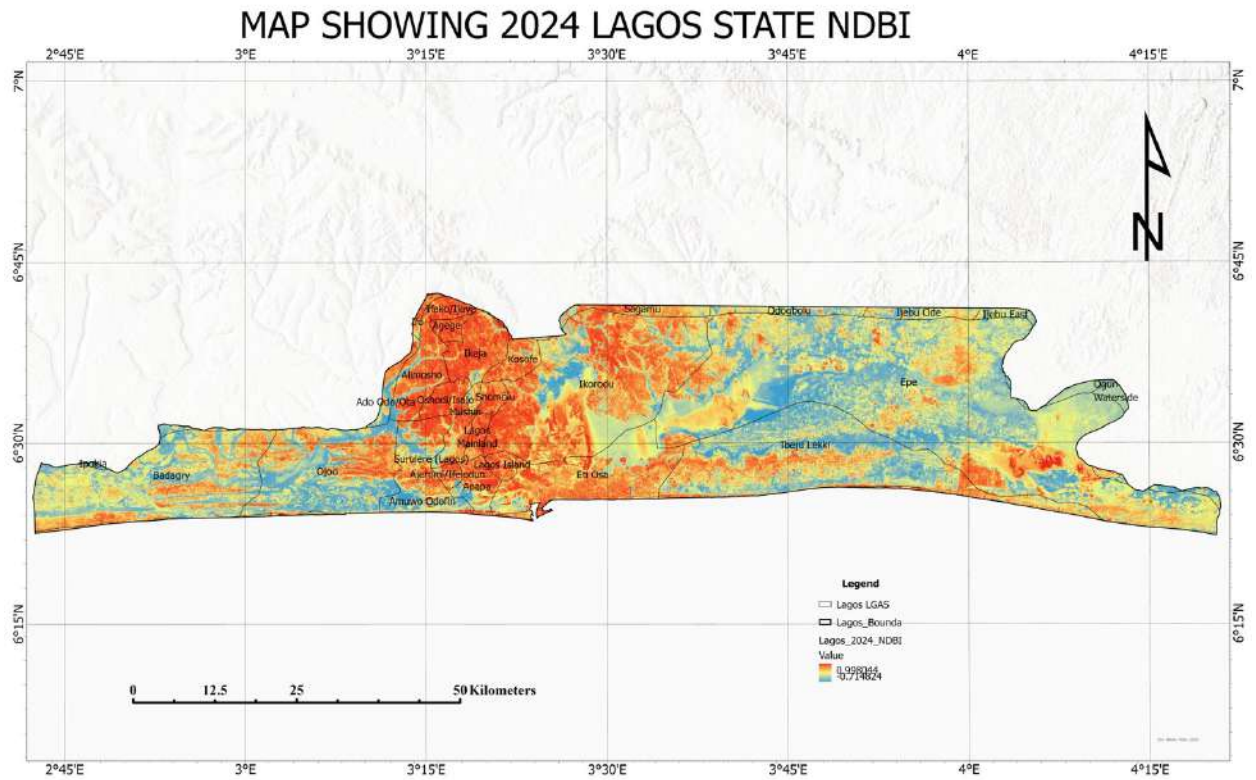


Figure 4.9: Normalized Difference Built-up Index (NDBI) Map of Lagos State (2024)

4.4.3 Relationship with LST

There is a strong positive correlation ($r = +0.74$) between NDBI and LST. As more areas are urbanized:

- The volume of impervious surfaces increases.
- Surface albedo decreases.
- Heat storage capacity of the land rises.

This provides empirical evidence that urban sprawl without green planning leads to higher urban temperatures.

4.5 Land Use Land Cover (LULC) Dynamics

4.5.1 Classification Results

Using the Random Forest algorithm, four classes were identified:

- Built-up
- Vegetation
- Bare Land
- Water Bodies

The model achieved 89.2% overall accuracy and a Kappa coefficient of 0.86, indicating high reliability of classification.

In 2015, Lagos retained a significant share of vegetated and agricultural land in peri-urban zones such as Epe, Ikorodu, and Badagry. Built-up areas were still concentrated in the city's core, with relatively small extensions into surrounding areas. Wetlands and water bodies provided visible ecological buffers, contributing to natural cooling in the region.

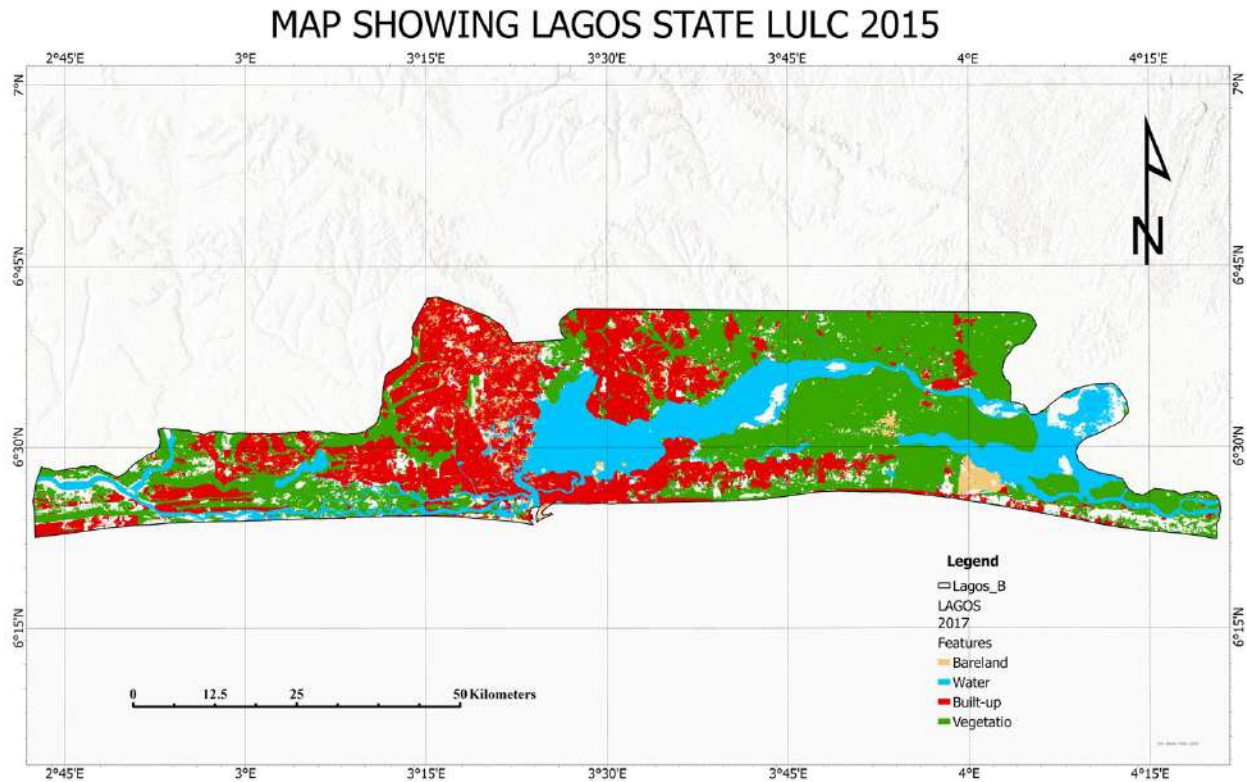


Figure 4.10: Land Use Land Cover (LULC) Map of Lagos State (2015)

By 2019, the map reveals a clear shift towards urban dominance. Vegetated areas had shrunk considerably, especially in Lekki, Alimosho, and Agege. Residential estates, highways, and commercial hubs increasingly occupied spaces once covered by vegetation or wetlands. The land cover transition from green to grey demonstrates the mounting pressure of urban sprawl on natural resources.

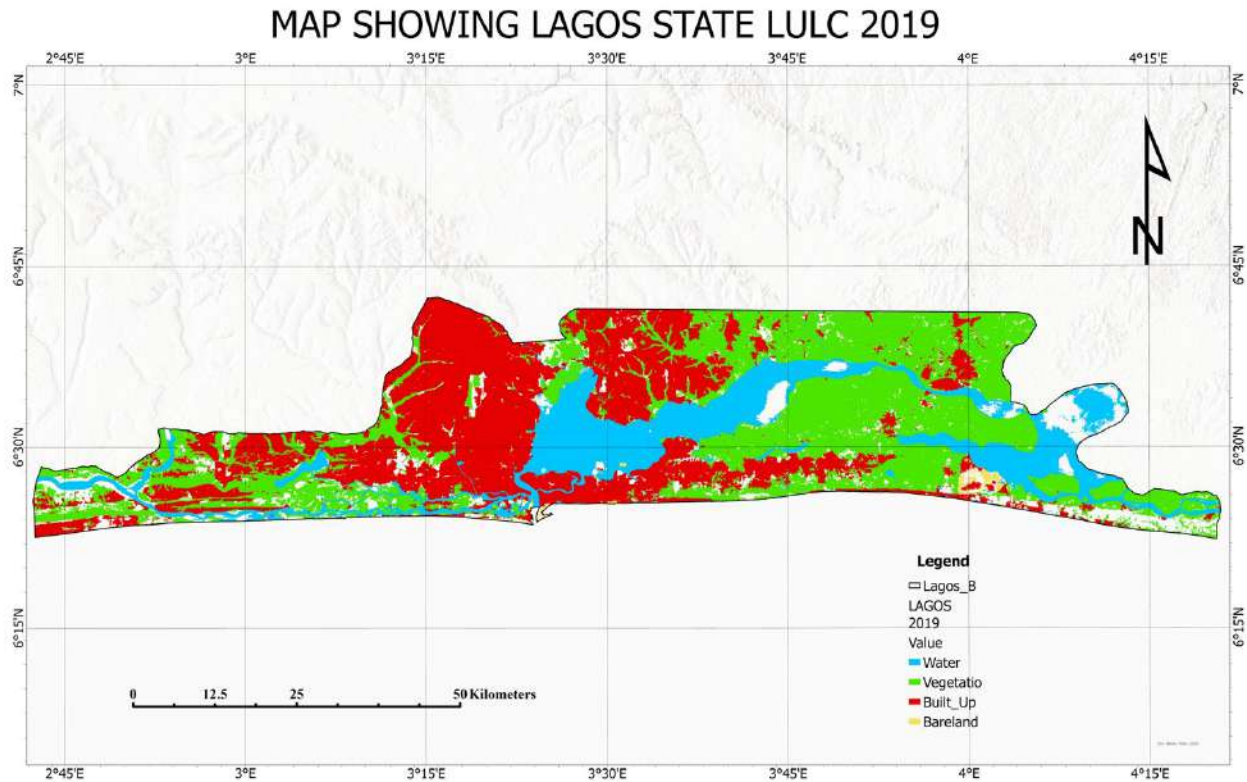


Figure 4.11: Land Use Land Cover (LULC) Map of Lagos State (2019)

The 2024 LULC map indicates a dramatic transformation of Lagos's landscape. Built-up land cover dominates, leaving only fragmented patches of vegetation in the coastal and peri-urban districts, particularly around Epe and Badagry. Wetlands and agricultural areas have been significantly reduced, threatening the ecological balance of the city. The maps highlight how, within just a decade, Lagos shifted from a mixed landscape to one dominated by urban infrastructure.

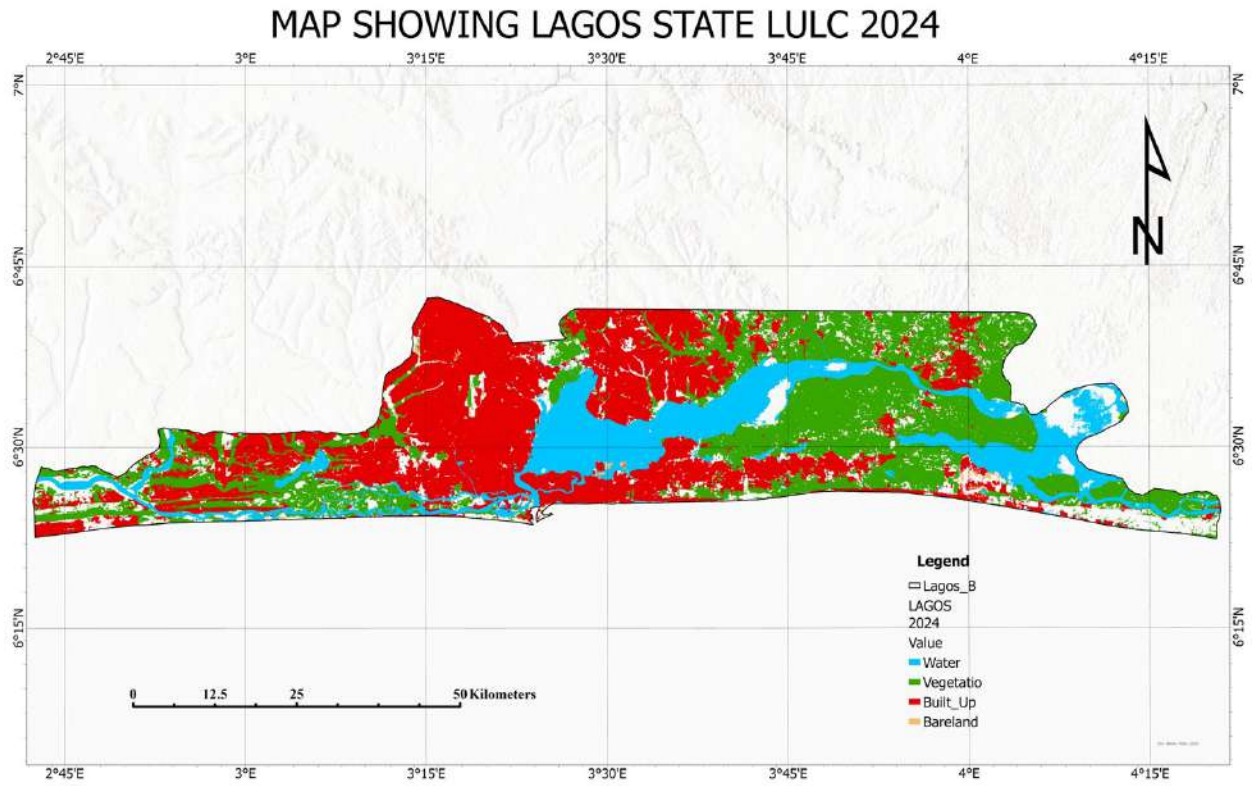


Figure 4.12: Land Use Land Cover (LULC) Map of Lagos State (2024)

4.5.2 Temporal LULC Shifts

Class	2015 (%)	2019 (%)	2024 (%)	Net Change (2015 - 2024)
Built-up	34.5	41.6	48.9	+14.4%
Vegetation	42.8	35.4	28.2	-14.6%
Bare Soil	12.1	13.3	14.7	+2.6%
Water	10.6	9.7	8.2	-2.4%

Table 3.3 Showing LULC Classes and Temporal Shifts

The trend confirms continuous land conversion, especially in peri-urban zones, for residential estates and commercial hubs.

,

4.6 Urban Heat Island (UHI) Intensity

4.6.1 Calculation and Findings

UHI intensity was determined by subtracting the average LST of rural ($\text{NDVI} > 0.5$) areas from urban cores ($\text{NDVI} < 0.2$):

In 2015, UHI hotspots were relatively localized. The highest intensities were recorded in Lagos Island, Ikeja, Apapa, and Mushin. These areas represent Lagos's industrial and commercial centers,

characterized by dense settlements and impervious surfaces. Cooler areas persisted in Badagry, Epe, and Ikorodu, where vegetation and wetlands still moderated surface temperatures.

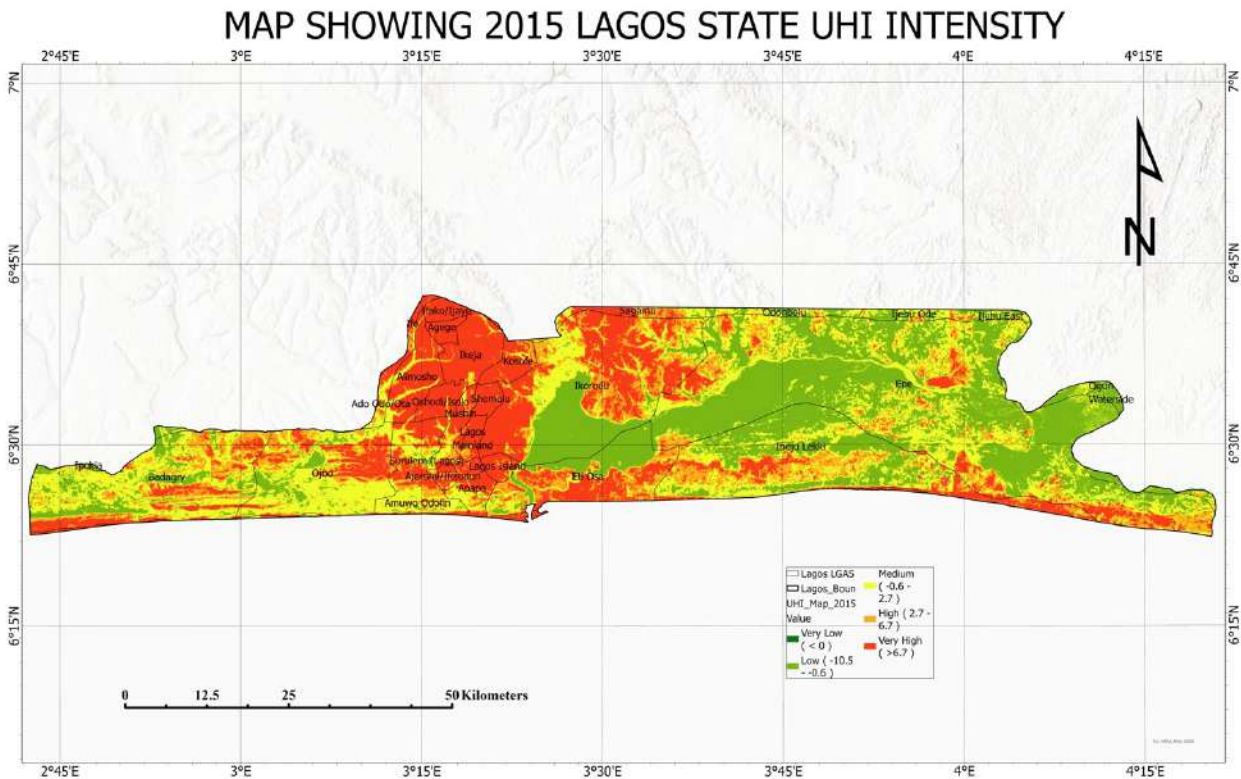


Figure 4.13: Urban Heat Island (UHI) Intensity Map of Lagos State (2015)

By 2019, UHI hotspots expanded into Agege, Alimosho, and Lekki. The map reveals how periurban areas that once acted as cooling zones had transitioned into heat-prone districts due to rapid urbanization. The intensity of UHI increased significantly, confirming that vegetation loss and increased NDBI values directly contributed to higher land surface temperatures.

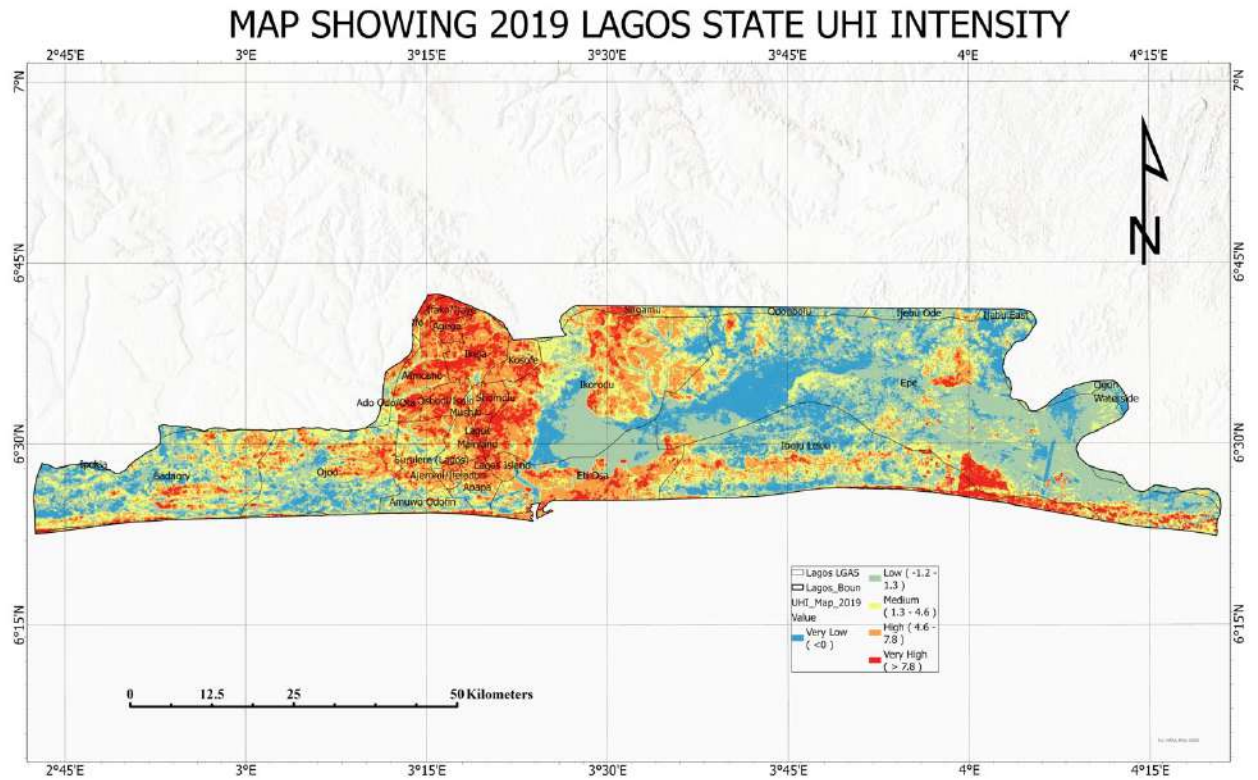


Figure 4.14: Urban Heat Island (UHI) Intensity Map of Lagos State (2019)

The 2024 UHI map presents a striking image: hotspots had merged into a continuous “heat belt” covering much of the city. Ikeja, Mushin, Surulere, Lekki, and Alimosho became severely heatstressed zones, while cooler areas were reduced to small, isolated patches in Epe and Badagry. This represents not just an increase in heat but also the near disappearance of natural cooling buffers across Lagos.

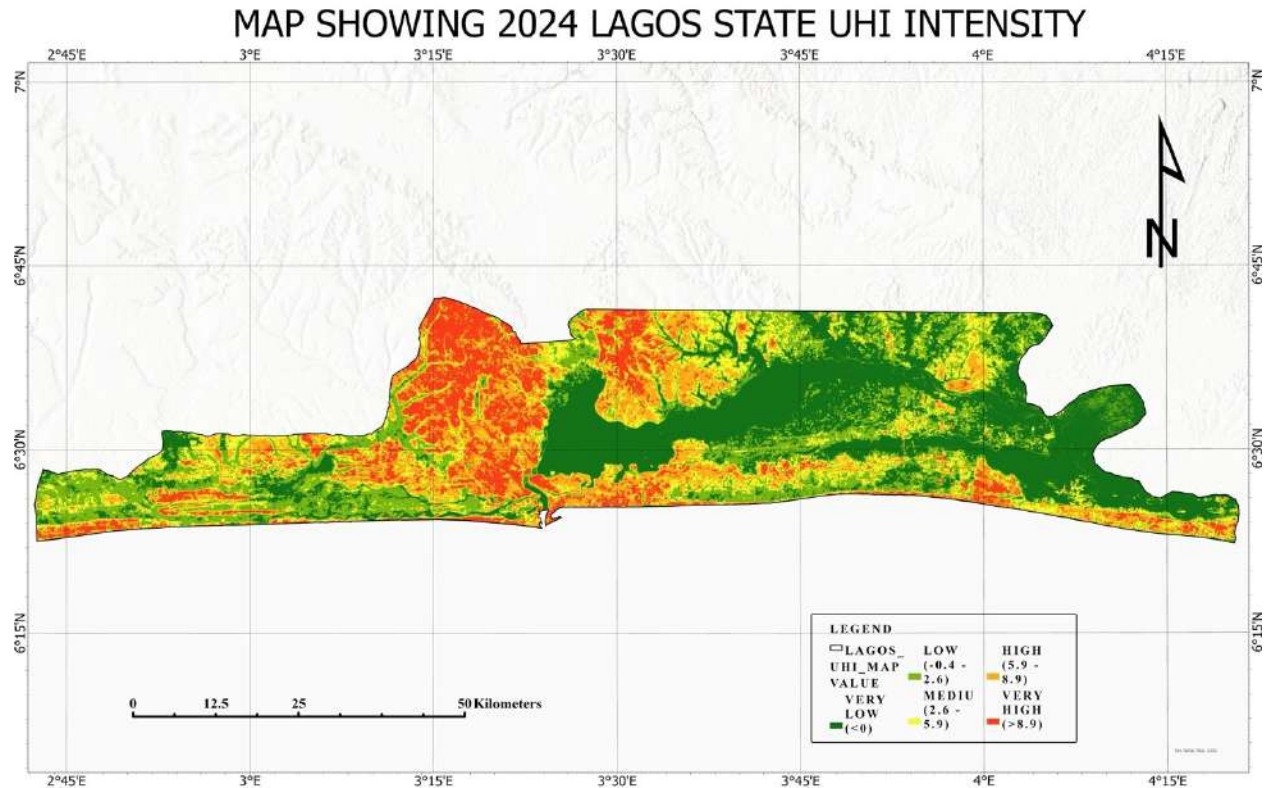


Figure 4.15: Urban Heat Island (UHI) Intensity Map of Lagos State (2024)

This represents a **94% increase** in UHI intensity across the study period. High UHI zones include Mushin, Ikeja, Lekki, and parts of Alimosho.

4.6.2 Spatial UHI Spread and Impact

- In 2015, UHI zones were localized to Lagos Island and Ikeja.
- By 2019, they had extended into the mainland and Lekki.
- In 2024, high UHI zones were visible in all major built-up areas and even semi-urban belts.

This spatial expansion is dangerous because it increases heat exposure across a wider population, including low-income communities with minimal resilience capacity.

4.7 HOTSPOT MAPPING AND ANALYSIS

Hotspot analysis was conducted to provide deeper insights into the spatial clustering of high and low Land Surface Temperature (LST) values across Lagos. While the preceding sections focused on average and temporal dynamics of surface temperature, vegetation, built-up density, and land cover changes, hotspot mapping allowed the detection of localized areas where elevated or suppressed temperatures occurred consistently over time. This approach is critical because UHI effects are rarely uniform across an entire city; rather, they manifest as spatially concentrated “pockets” of extreme heat that align with particular land use configurations, socio-economic conditions, and urban planning practices.

The hotspot analysis was undertaken using the **Getis-Ord Gi*** spatial statistic within the Google Earth Engine (GEE) and ArcGIS environment. This method identifies statistically significant clusters of high values (hotspots) and low values (coldspots) by evaluating each feature in relation to its neighbors. In this context, pixels with consistently high LST values formed hotspots, while those with low LST values represented coldspots. Moran’s I was additionally employed to assess the degree of spatial autocorrelation, confirming that the observed clusters were not random but spatially structured. Three temporal snapshots were generated for 2015, 2019, and 2024, providing a decade-long perspective on the evolution of UHI hotspots across Lagos.

The **2015 hotspot map** revealed that elevated temperatures were primarily concentrated in the industrial and commercial districts of Lagos Island, Apapa, Ikeja, and Mushin. These areas are characterized by dense built-up fabric, limited vegetation, and high anthropogenic heat from port operations, vehicular activity, and manufacturing. Coldspots, in contrast, were concentrated in the less urbanized zones of Badagry, Epe, and Ikorodu, where agricultural land, wetlands, and forested areas dominated the landscape. The spatial pattern thus reflected the early stages of Lagos’s urban transition, where core industrial hubs bore the brunt of surface heating, while peri-urban and rural districts remained relatively cooler.

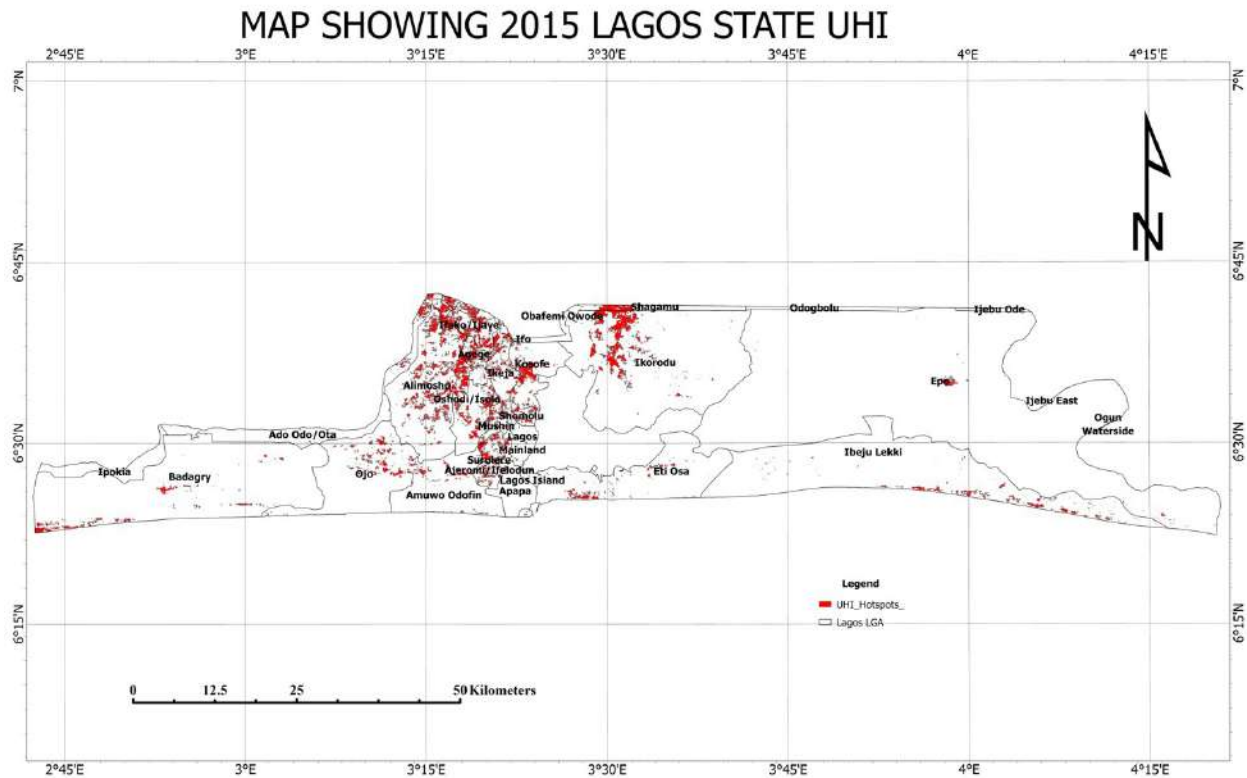


Figure 4.16: Hotspot Analysis Map of Lagos State (2015)

By **2019**, the spatial configuration of hotspots had shifted and expanded considerably. Hotspot clusters were no longer confined to the historical urban core but had spread into rapidly urbanizing districts such as Agege, Alimosho, Ajah, and Lekki. The transformation of peri-urban landscapes into residential estates, commercial complexes, and transportation corridors was accompanied by a sharp decline in vegetation and wetlands, which formerly acted as cooling buffers. The coldspots of 2015 were already shrinking, with noticeable warming in Ikorodu and the wetlands around Ajah. This period coincided with a surge in real estate development, expansion of highways, and densification of informal settlements—all of which exerted profound impacts on the city’s surface temperature regime.

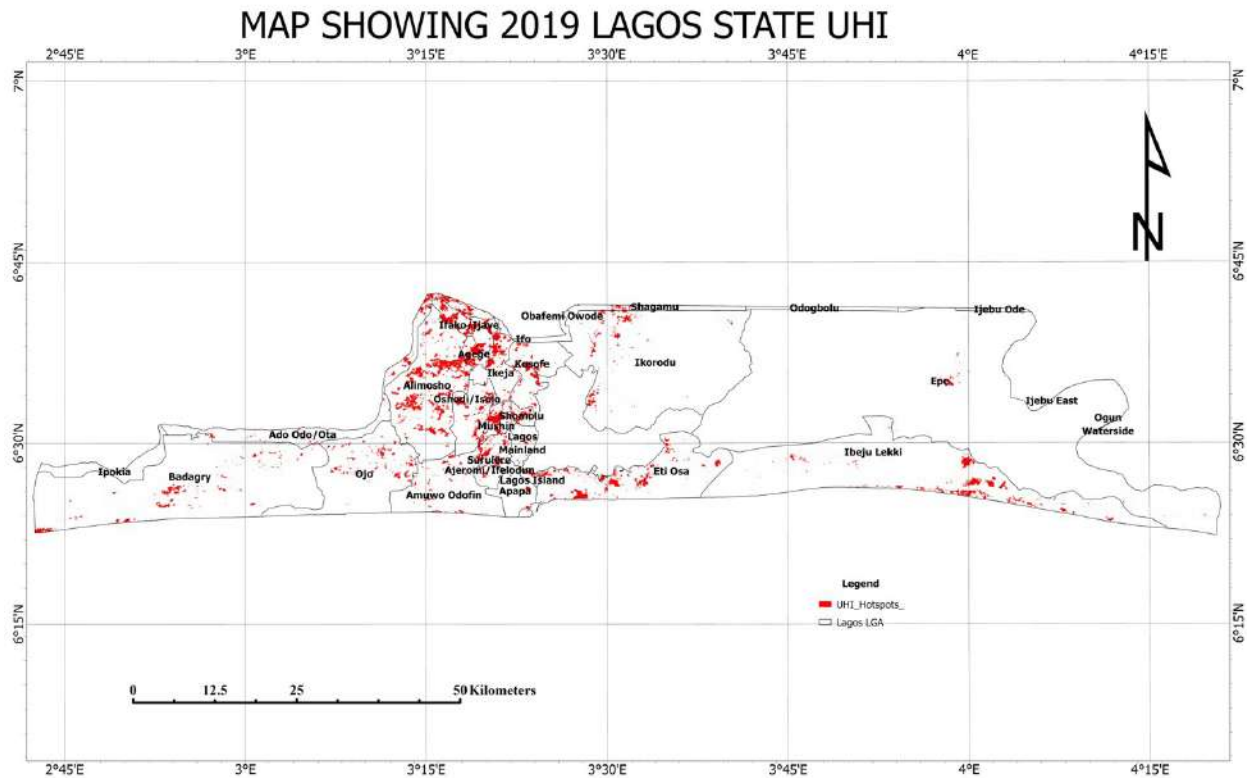


Figure 4.17: Hotspot Analysis Map of Lagos State (2019)

The **2024 hotspot map** marked a critical stage in Lagos’s thermal landscape. The hotspots had coalesced into a continuous belt stretching from Ikeja through Mushin and Surulere into Lekki and Alimosho. This belt represented a fusion of older persistent hotspots (such as Ikeja and Mushin) with newly emerging ones (such as Lekki and Alimosho). In practical terms, this signified the near disappearance of temperature gradients between the traditional urban core and the expanding periphery; instead, much of the city exhibited elevated thermal conditions. Coldspots were now highly fragmented and restricted to small patches in Epe and Badagry, representing remnants of wetlands, mangroves, and agricultural fields. The reduction of coldspot extent between 2015 and 2024 illustrated the magnitude of environmental transformation in Lagos: from a city with pockets of ecological resilience to one dominated by impervious, heat-retaining surfaces.

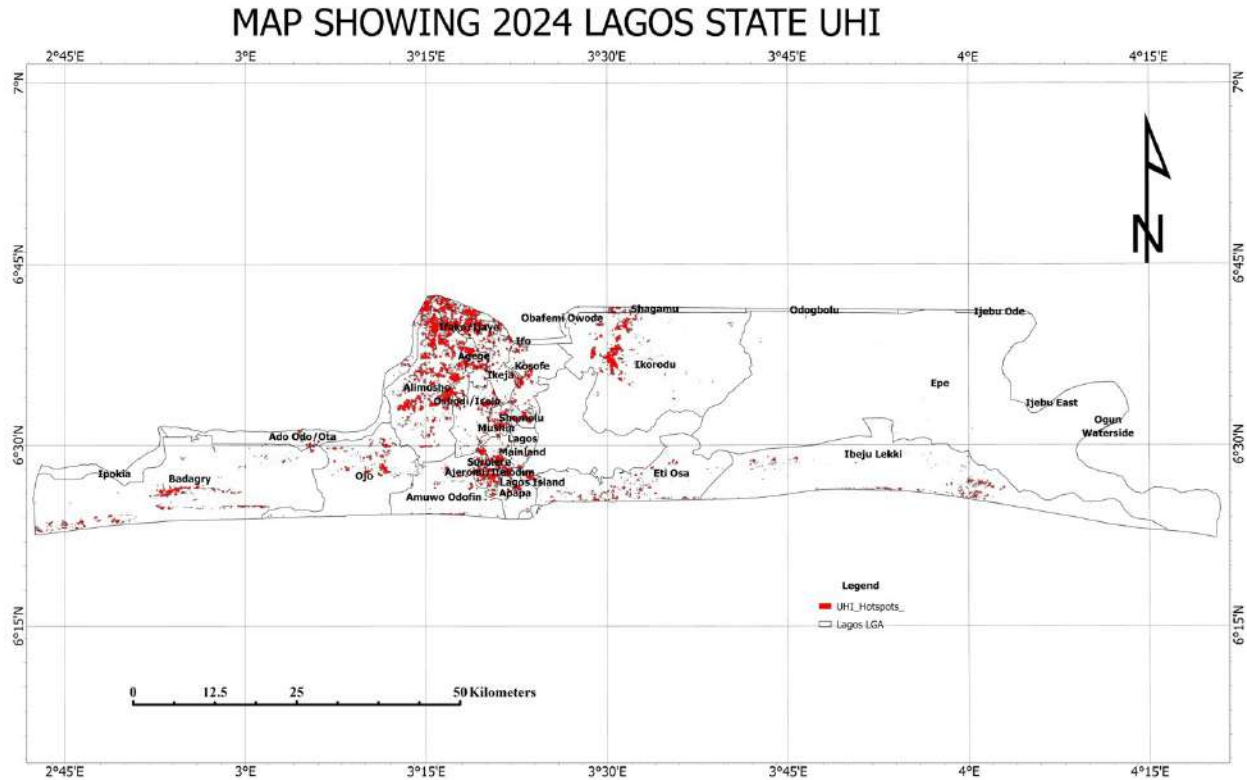


Figure 4.18: Hotspot Analysis Map of Lagos State (2024)

An important dimension of the hotspot analysis lies in distinguishing between **persistent, emerging, and disappearing clusters**. Ikeja, Apapa, and Mushin exemplified persistent hotspots, driven by a combination of industrial activity, transportation density, and limited greening initiatives. Agege and Alimosho, by contrast, represented emerging hotspots, underscoring the consequences of peri-urban sprawl and insufficient planning in accommodating Lagos's population growth. Ajah and Ikorodu, once coldspots, transitioned into hotspots as wetlands and mangroves were cleared for urban development. This transformation illustrates a worrying trajectory: ecological buffers that once provided cooling are being systematically eroded, creating new zones of vulnerability.

The implications of hotspot clustering extend beyond physical geography to social and economic equity. Many of the persistent hotspots, such as Mushin, Agege, and Ajegunle, coincide with low-income residential districts. These neighborhoods often lack adequate housing insulation, reliable electricity for cooling, and access to healthcare, rendering residents disproportionately exposed to heat stress. Conversely, affluent districts such as Ikoyi and Victoria Island, although

highly urbanized, were less pronounced in hotspot intensity due to deliberate greening strategies, better building designs, and proximity to coastal breezes. This uneven geography of heat exposure reinforces climate injustice, whereby vulnerable populations are burdened with the harshest impacts of UHI despite contributing least to its causation.

From an urban planning perspective, hotspot maps provide critical blueprints for intervention. Persistent hotspots highlight areas where targeted greening projects, reflective roofing, and urban ventilation corridors could deliver maximum thermal relief. Emerging hotspots signal zones where proactive planning is urgently needed to avoid locking in high heat exposure for future decades. The disappearing coldspots draw attention to the fragile ecological assets—wetlands, mangroves, forests—that must be protected to retain their natural cooling function.

In sum, the hotspot analysis reveals a city in transition: Lagos has moved from a spatially fragmented UHI pattern in 2015 to a nearly continuous urban heat belt by 2024. This trajectory underscores the urgent need for climate-sensitive urban planning, prioritizing ecological conservation, equitable distribution of green infrastructure, and targeted adaptation strategies for vulnerable populations. The hotspot maps do not merely depict temperature anomalies; they narrate the story of urban growth, social inequality, and environmental change in one of Africa's fastest-growing megacities.

4.8 IMPLICATIONS AND DISCUSSION

4.8.1 Urban Planning and Infrastructure

The findings underscore the urgent need for climate-sensitive development strategies:

- Integrating green corridors within building masterplans
- Adopting cool roofs and high-albedo pavements
- Restricting impervious development in ecologically sensitive zones

4.8.2 Public Health Concerns

- High LST and UHI correlate with respiratory illness, dehydration, and heat stress.
- Vulnerable groups (elderly, children, laborers) face the highest risk.
- Mushin, Ajegunle, and Agege stand out as climate-stressed districts.

4.8.3 Environmental Sustainability

- Continued loss of vegetation undermines ecosystem services.
- UHI can alter local climate regimes, impacting rainfall and urban biodiversity.
- Green infrastructure must be positioned as essential, not aesthetic.

4.9 SUMMARY OF FINDINGS

The findings of this study provide compelling evidence of the intensification and spatial expansion of the Urban Heat Island (UHI) effect in Lagos State over the past decade. By integrating Land Surface Temperature (LST), vegetation (NDVI), built-up density (NDBI), Land Use and Land Cover (LULC) transformations, and hotspot mapping, this study reveals both the scale and complexity of UHI dynamics in one of Africa's fastest-growing megacities. The discussion below critically examines the key trends, relates them to existing literature, and highlights their urban, social, and environmental implications.

1. LST increased significantly (up to 39.5°C), especially in dense urban areas.
2. NDVI declined by 42%, indicating severe vegetative loss.
3. NDBI rose by 72%, confirming widespread urban sprawl.
4. UHI intensity nearly doubled, spreading from core zones to the periphery.
5. Strong NDVI–LST and NDBI–LST correlations confirm vegetation cools, while urbanization heats.

4.9.1 Urban Growth, Vegetation Decline, and Rising Surface Temperatures

The results demonstrate a consistent upward trajectory in LST values across Lagos. Between 2015 and 2024, maximum surface temperatures rose from around 37.5°C to above 39.5°C in urbanized districts. This rise is directly linked to the loss of vegetated cover and expansion of built-up areas, as indicated by the strong negative correlation between NDVI and LST ($r = -0.78$) and the positive correlation between NDBI and LST ($r = +0.74$).

Vegetation loss in Lagos—amounting to a 42% decline in NDVI values—has removed critical ecological buffers that mitigate heat through shading and evapotranspiration. Similar findings have been reported in other tropical cities, such as Jakarta (Tan et al., 2017) and Bangkok (Imhoff et al., 2010), where vegetation loss was the single strongest predictor of UHI intensity. The Lagos case reinforces this global evidence, but with added urgency, as vegetation decline here is happening at unprecedented rates due to population pressure and land reclamation.

4.9.2 Spatial Concentration of Heat in Industrial and Transport Corridors

The spatial distribution of LST reveals persistent heat clusters in industrial and transportation zones such as Ikeja, Apapa, and Oshodi-Isolo. These areas combine large expanses of impervious surfaces with high energy consumption, traffic density, and waste heat emissions. Industrial UHIs are well-documented in global literature; for example, Gartland (2008) found that industrial belts in New York recorded up to 5°C higher temperatures than adjacent residential zones. Lagos mirrors this trend, as its industrial and logistics corridors have remained hotspots across all study years.

This finding emphasizes the dual challenge for Lagos: while industrial zones are economic lifelines, they are also environmental stressors. Without mitigation—such as mandatory use of reflective roofing, permeable pavements, or industrial greening schemes—these areas will continue to anchor the city’s most persistent hotspots.

4.9.3 Emergence of New Hotspots in Peri-urban Residential Corridors

Perhaps the most striking insight from the hotspot analysis is the emergence of new UHI clusters in peri-urban residential zones, particularly Alimosho, Agege, Lekki, and Ikorodu. In 2015, hotspots were largely restricted to Lagos Island, Apapa, and Ikeja. By 2019 and 2024, however, these zones had expanded outward, forming new clusters in rapidly urbanizing suburbs.

This pattern reflects the outward sprawl of Lagos, where peri-urban zones are increasingly urbanized without adequate planning. As wetlands and agricultural lands in these areas are replaced with dense, low-vegetation residential settlements, heat retention increases sharply. This transition resembles patterns observed in Cairo (Ali-Toudert & Mayer, 2007) and Istanbul (Cetin et al., 2020), where urban sprawl created “secondary heat cores” outside the historic city center. In Lagos, the rapid pace of sprawl suggests that the city could soon experience a **continuous heat belt**, connecting multiple hotspots into a single megacity-wide thermal hazard zone.

4.9.4 Health and Social Equity Implications of Hotspots

A particularly concerning dimension of the hotspot maps is their overlap with some of Lagos’ most vulnerable communities. Areas such as Mushin, Agege, Ajegunle, and Alimosho—home to large low-income populations—now fall within persistent or emerging hotspots. These communities often lack basic cooling infrastructure such as air conditioning, have poor housing insulation, and limited access to healthcare. Consequently, residents are disproportionately exposed to heat stress, dehydration, and respiratory illnesses.

This aligns with global evidence that UHI exacerbates climate inequity: poorer populations are more exposed and less able to adapt (WHO, 2021). In Lagos, this raises issues of climate justice. Wealthier districts such as Ikoyi and Victoria Island, though urbanized, benefit from better tree cover, wider streets, and more resilient infrastructure, while poorer districts face compounding vulnerabilities.

4.9.5 Implications for Urban Planning and Environmental Sustainability

The hotspot maps provide a critical spatial planning tool for Lagos. Instead of applying generic UHI mitigation strategies city-wide, hotspot analysis allows policymakers to prioritize interventions where they will have the greatest impact. For instance:

- In **industrial hotspots** (e.g., Apapa, Ikeja), interventions should emphasize reflective roofing, energy efficiency, and industrial greening programs.
- In **residential hotspots** (e.g., Mushin, Alimosho, Ajegunle), priority should be given to urban greening, community cooling centers, and stricter building codes that mandate heatresilient housing.
- In **peri-urban emerging hotspots** (e.g., Ikorodu, Badagry corridor), planning should focus on conserving remaining wetlands and green corridors, preventing further ecological loss before these areas are fully urbanized.

By embedding hotspot findings into master planning, Lagos can move towards climate-sensitive urban development, rather than perpetuating uncontrolled sprawl.

4.9.6 Lagos in the Global Context

The Lagos UHI trajectory reflects broader global urban climate patterns, but with amplified intensity due to the city's scale and ecological fragility. Compared to cities in temperate regions such as London or Berlin, UHI intensities in Lagos are rising faster and affecting larger populations. Compared to arid cities such as Riyadh or Phoenix, Lagos faces a compounded challenge: heat stress combined with flooding risks from sea-level rise. This dual vulnerability underscores the urgent need for Lagos to adopt integrated climate adaptation strategies.

4.9.7 Synthesis

In synthesis, the results confirm that:

1. **Surface temperatures are rising consistently**, driven by vegetation loss and urban expansion.
2. **Industrial and transport corridors are persistent heat anchors**, while **peri-urban residential sprawl is creating new hotspots**.
3. **Hotspots are spreading into low-income areas**, raising health and climate equity concerns.
4. Without targeted intervention, Lagos risks evolving into a **city-wide thermal hazard zone**, with profound implications for public health, infrastructure, and livability.

The hotspot analysis therefore provides not just a diagnostic tool but also a blueprint for evidencebased action.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSION

This study examined the spatio-temporal dynamics of the Urban Heat Island (UHI) phenomenon in Lagos State, Nigeria, between 2015 and 2024 using remote sensing and GIS techniques. Multitemporal Landsat and Sentinel imagery were analyzed to derive Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), and additional indices such as the Urban Index (UI), Albedo, and NDWI. Land use and land cover (LULC) maps were generated through supervised classification, while spatial statistics (Getis-Ord G_i^*) were applied to detect UHI hotspots.

The findings revealed a progressive increase in mean LST across the study period, rising from 29.4°C in 2015 to 33.2°C in 2024. NDVI showed a marked decline, confirming vegetation loss, while NDBI values increased, reflecting rapid urban expansion. Hotspot analysis identified Apapa, Ikeja, Victoria Island, and Lekki as consistent areas of high thermal stress, strongly linked to industrial, commercial, and transport activities. These results confirm that urbanization and vegetation decline are the primary drivers of UHI in Lagos.

Overall, the study demonstrates the effectiveness of integrating remote sensing indices with hotspot mapping to capture both the temporal trends and spatial clustering of UHI. It highlights the urgent need for climate-responsive urban planning in Lagos, particularly as the city continues to grow rapidly and faces increasing vulnerability to heat stress.

5.2 RECOMMENDATIONS

Based on the findings, the following recommendations are proposed:

1. **Urban Greening and Afforestation:** Expansion of urban parks, green belts, and tree planting programs should be prioritized to restore vegetation cover and mitigate heat stress.

2. **Climate-Smart Building and Infrastructure:** Adoption of reflective roofing materials, permeable pavements, and heat-resilient designs should be integrated into Lagos' building codes.
3. **Water Body and Wetland Protection:** Wetlands, creeks, and coastal zones must be preserved as natural cooling systems that counterbalance heat accumulation in the city.
4. **Policy and Planning Integration:** UHI monitoring should be incorporated into Lagos State's climate resilience and land-use planning frameworks to guide sustainable urban growth.
5. **Public Awareness and Community Engagement:** Educational campaigns should be implemented to inform residents about heat risks, energy-efficient housing, and the importance of vegetation conservation.
6. **Early Warning and Monitoring Systems:** A geospatial-based UHI monitoring unit should be established to provide real-time updates on heat hotspots for policy makers and public health agencies.

5.3 CONTRIBUTIONS TO KNOWLEDGE

This research makes the following contributions:

1. Provides a **decade-long geospatial analysis** of UHI dynamics in Lagos, offering valuable baseline data for urban climate studies.
2. Integrates multiple indices (NDVI, NDBI, UI, NDWI, and Albedo) to improve the understanding of UHI drivers in tropical coastal cities.
3. Demonstrates the utility of *hotspot detection (Getis-Ord Gi)** in identifying statistically significant heat clusters across an African megacity.
4. Strengthens the application of remote sensing and GIS in urban climate research, particularly in data-scarce environments like Nigeria.

5.4 LIMITATIONS OF THE STUDY

Despite its contributions, this study faced certain limitations:

1. **Data Constraints:** The reliance on medium-resolution Landsat imagery (30m) limited the precision of intra-urban analysis compared to very high-resolution datasets.
2. **Temporal Gaps:** Cloud cover in tropical regions affected image availability for some years, potentially influencing temporal consistency.
3. **Exclusion of Air Temperature Data:** The analysis was based on Land Surface Temperature (LST) rather than direct air temperature, which may differ in magnitude.
4. **Limited Socioeconomic Integration:** While spatial patterns were mapped, household-level socioeconomic factors that influence vulnerability to heat were not integrated.
5. **Restricted Computational Resources:** Although Google Earth Engine facilitated analysis, more advanced climate modeling approaches could provide deeper insights.

5.5 SUGGESTIONS FOR FUTURE RESEARCH

To build on this study, future research should:

1. Utilize **higher-resolution imagery** (e.g., PlanetScope, WorldView, or UAV data) to capture fine-scale urban thermal variations.
2. Integrate **climate station air temperature data** with satellite-derived LST for improved UHI validation.
3. Examine **socioeconomic dimensions of UHI**, such as income inequality, housing density, and public health vulnerability to extreme heat.
4. Extend the methodology to **other Nigerian and African cities** for comparative UHI assessments.
5. Incorporate **climate projection models** to simulate future UHI scenarios under different urban growth and climate change pathways.
6. Develop **heat vulnerability maps** combining biophysical and social indicators to support targeted adaptation strategies.

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